

Project Management

8. Risk Management & Change Control

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- Plan Risk Responses
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Planning for the unknown

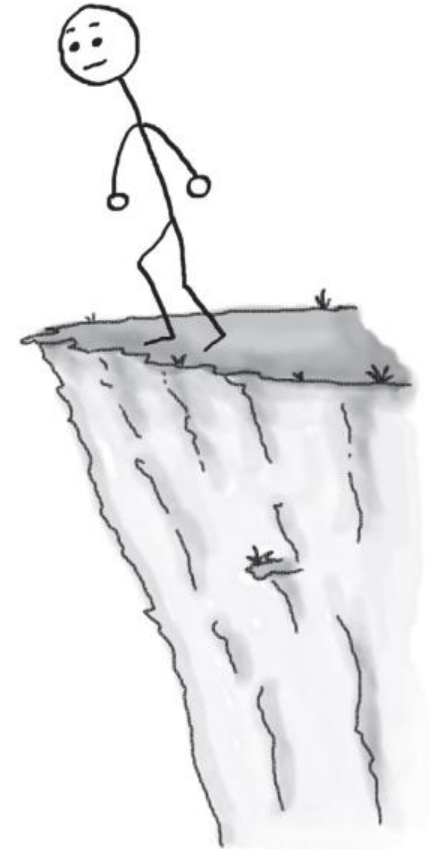


Risk Management

Even the most carefully planned project can run into trouble.

What is Risk

- A risk is any ***uncertain event or condition*** that might affect your project.
 - **Event risk** is specific uncertain occurrence that could potentially impact the project. (*Ex. The lead developer might resign during the project.*)
 - **Condition risk** refers to a known or ongoing state or condition that could lead to problems in the future. (*Ex. The team is unfamiliar with the new frontend framework selected for the project.*)
- Risk may be negative or positive
 - **Negative risks** (threats): These can harm your project. (*Ex. Vendor delay in delivering critical API integration*)
 - **Positive risks** (opportunities): These can benefit your project if managed properly. (*Ex. New AI tool may accelerate test cycles and reduce workload*)



Opportunities can be risky

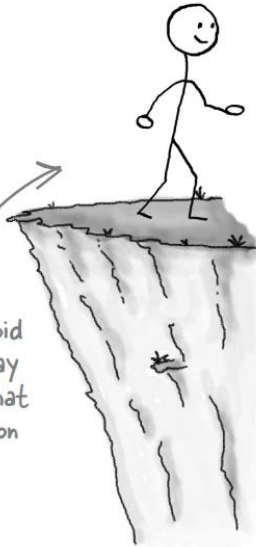


How you deal with risk?

1 Avoid

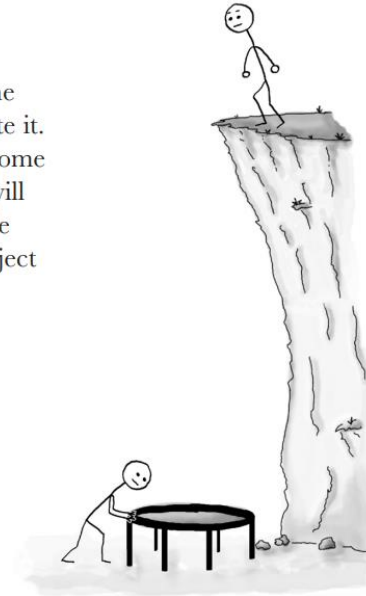
The best thing that you can do with a risk is avoid it—if you can prevent it from happening, it definitely won't hurt your project.

The easiest way to avoid this risk is to walk away from the cliff...but that may not be an option on this project.



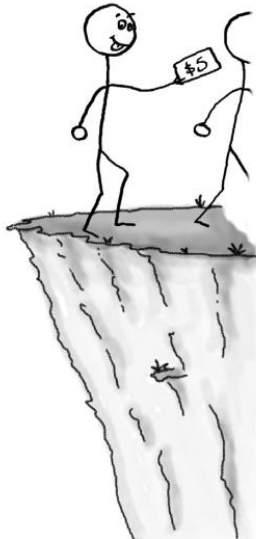
2 Mitigate

If you can't avoid the risk, you can mitigate it. This means taking some sort of action that will cause it to do as little damage to your project as possible.



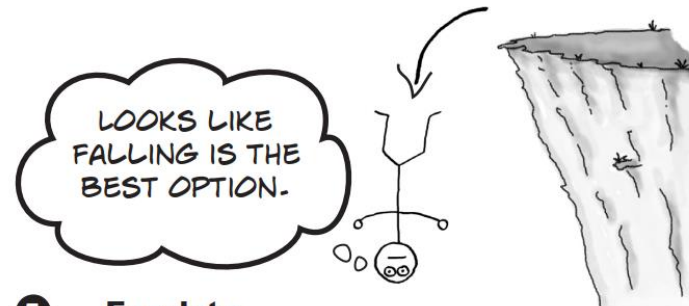
3 Transfer

One effective way to deal with a risk is to pay someone else to accept it for you. The most common way to do this is to buy insurance.



4 Accept

When you can't avoid, mitigate, or transfer a risk, then you have to accept it. But even when you accept a risk, at least you've looked at the alternatives and you know what will happen if it occurs.



5 Escalate

If the risk is not in your project's scope, you might need to tell somebody else about it to find an appropriate response.

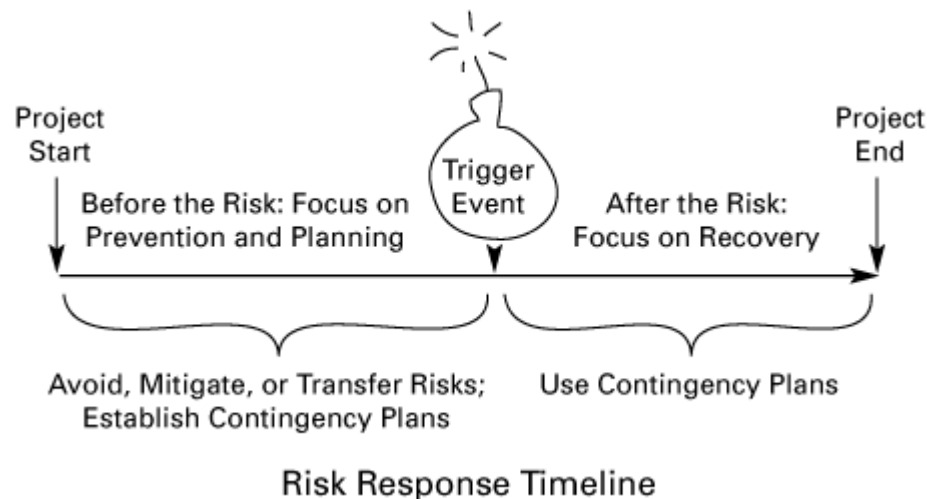
Related terms

Risk Trigger

A risk trigger is an event or situation that indicates that a **risk is about to occur or has occurred**. We can think of risk triggers as warning signs or risk symptoms.

Issue

An issue is **a risk that has occurred**. It is no longer a possibility of harm, loss or gain, so we cannot call it a risk anymore. Instead, it has converted from being a risk to being an issue. Issues are often recorded in the project Issue log.



Individual risk	Overall project risk
Individual project risk is an uncertain event or condition that, if it occurs, has a positive or negative effect on one or more project objectives.	Overall project risk is the effect of uncertainty on the project as a whole , arising from all sources of uncertainty including individual risks, both positive and negative.
What are the risks in my project?	How risky is my project?
Specific risks within the project	The overall riskiness of the project
Recorded in a Risk Register or similar document.	Result in project selection and Risk Reports
From the project manager's perspective	From the project sponsor's perspective
Mostly during the remainder of the project lifecycle.	During the pre-project or concept phase

Project Risk Management considers both individual project risks and sources of overall project risk.

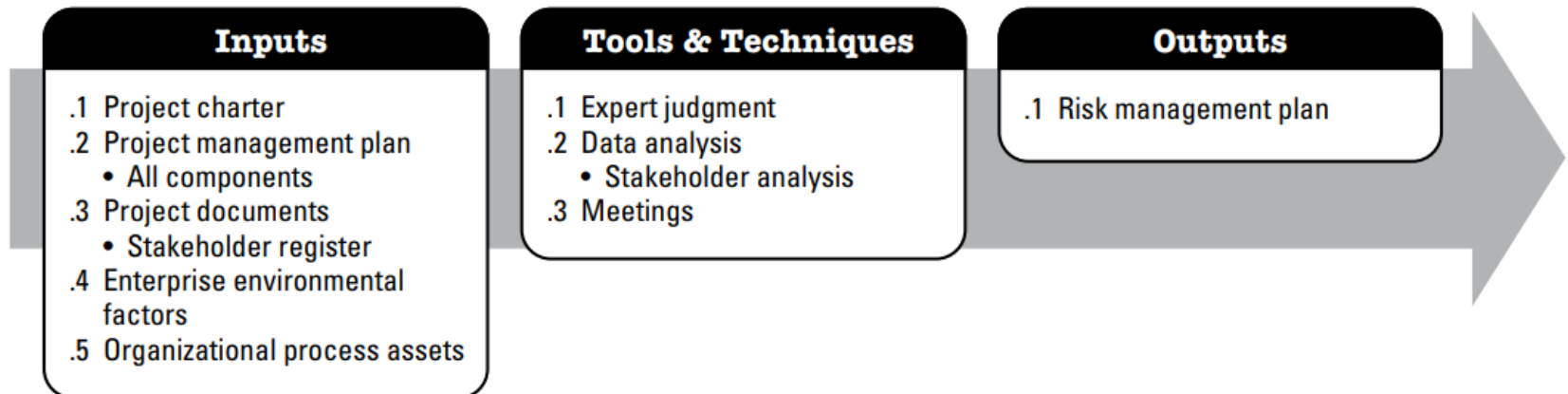
Risk Management Process



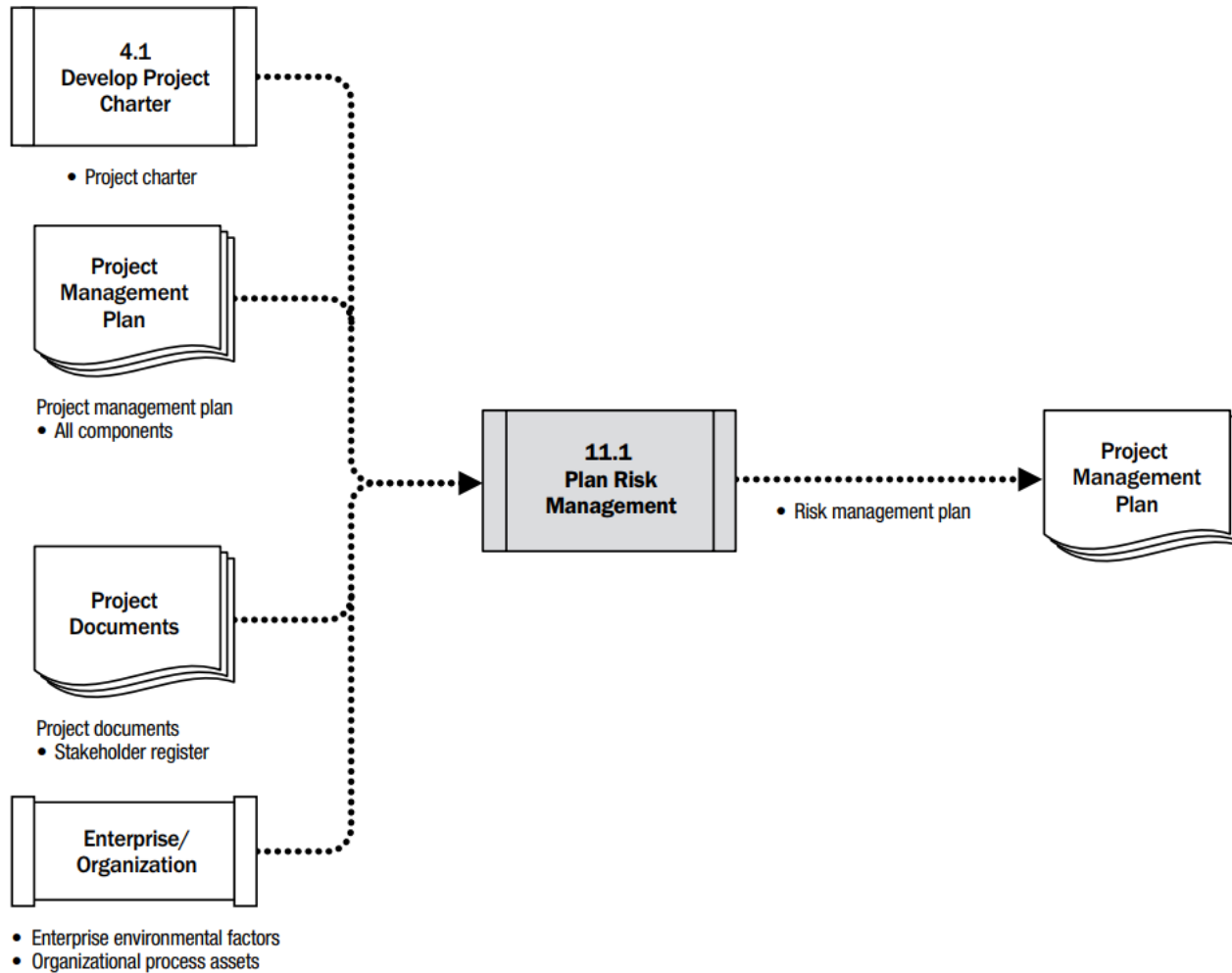
Plan Risk Management

Plan Risk Management

- **What:** process of defining how to conduct risk management activities for a project
- **Why:** ensures that the degree, type, and visibility of risk management are proportionate to both risks and the importance of the project to the organization and other stakeholders.
- **When:** performed once or at predefined points in the project.
 - Should begin when a project is conceived and should be completed early in the project.
 - May revisit later, for example at a major phase change, or if the project scope changes significantly.



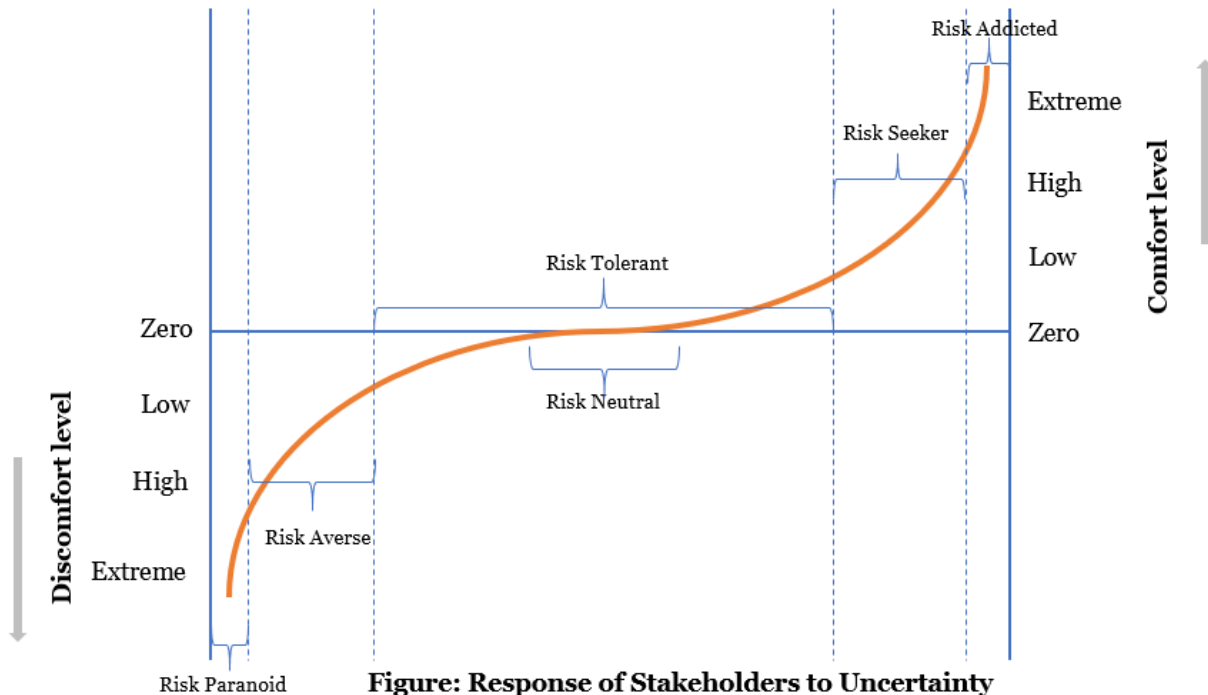
Data Flow



Tools & Techniques

Stakeholder analysis

- This is useful in determining stakeholders' roles and responsibilities for managing risk on the project, as well as their **risk attitudes**.
- A person can be both risk averse and risk seeking at different times.



Outputs: Risk Management plan

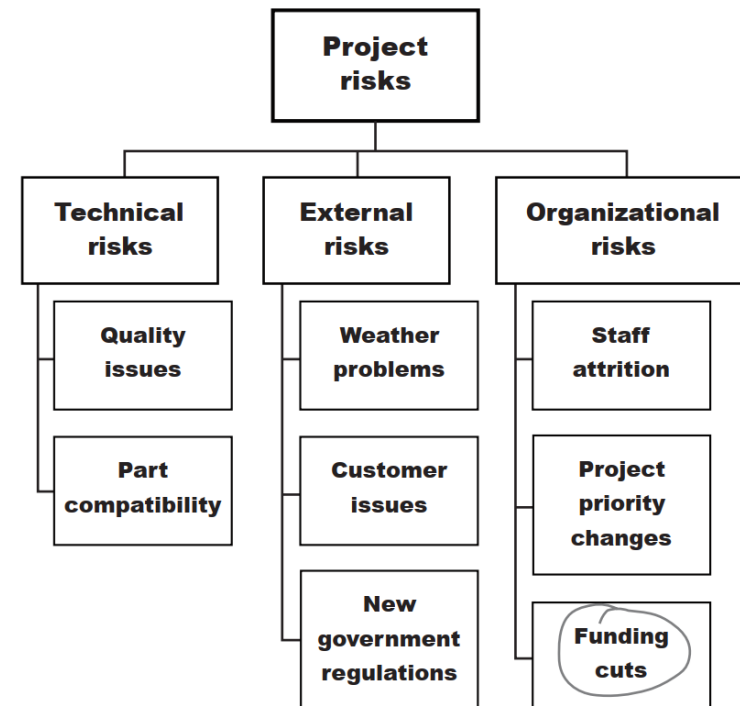
- Describes how risk management activities will be structured and performed
 - Risk strategy
 - Methodology: specific approaches, tools, and data sources
 - Roles and responsibilities.
 - Funding: funds needed to perform activities related to risk mngt.
 - Timing: when and how often the processes will be performed
 - Risk categories: Risk breakdown structure (RBS)
 - Stakeholder risk appetite: should be expressed as measurable risk thresholds .
 - Definitions of risk probability and impacts.
 - Probability and impact matrix.
 - Reporting formats.
 - Tracking.



Risk Management plan

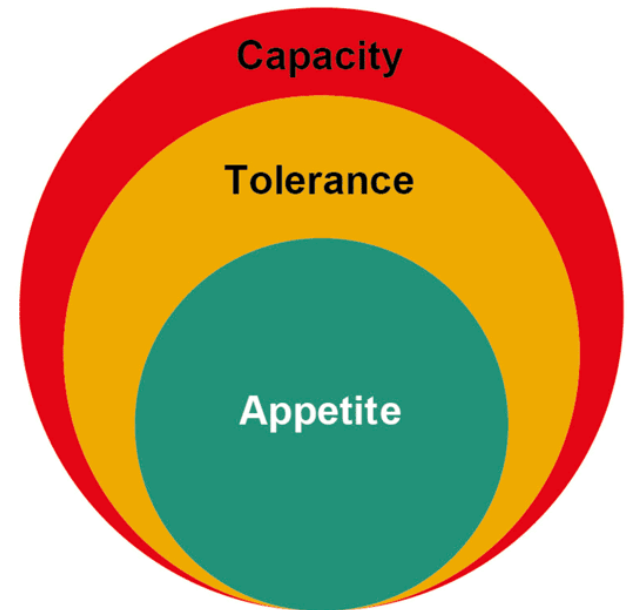
Risk Breakdown Structure (RBS)

RBS LEVEL 0	RBS LEVEL 1	RBS LEVEL 2
0. ALL SOURCES OF PROJECT RISK	1. TECHNICAL RISK	1.1 Scope definition
		1.2 Requirements definition
		1.3 Estimates, assumptions, and constraints
		1.4 Technical processes
		1.5 Technology
		1.6 Technical interfaces
		Etc.
	2. MANAGEMENT RISK	2.1 Project management
		2.2 Program/portfolio management
		2.3 Operations management
		2.4 Organization
		2.5 Resourcing
		2.6 Communication
		Etc.
	3. COMMERCIAL RISK	3.1 Contractual terms and conditions
		3.2 Internal procurement
		3.3 Suppliers and vendors
		3.4 Subcontracts
		3.5 Client/customer stability
		3.6 Partnerships and joint ventures
		Etc.
	4. EXTERNAL RISK	4.1 Legislation
		4.2 Exchange rates
		4.3 Site/facilities
		4.4 Environmental/weather
		4.5 Competition
		4.6 Regulatory
		Etc.



Stakeholders risk appetite

Risk appetite	The amount and type of risk that an organisation is willing to pursue or retain.
Risk tolerance	The acceptable degree of variability, or deviation from the expected level of risk that an organisation is prepared to withstand , in order to achieve its objectives.
Risk capacity	The maximum level of risk to which the organisation should/can be exposed.



Risk probability and impacts

Risk Probability

The likelihood that a specific risk will occur.

- Scale: Typically expressed qualitatively (e.g., Low, Medium, High) or quantitatively (e.g., 0% to 100%).
- Example: "There is a 70% chance that supplier delay will occur."

Risk Impact

The **extent of consequences** or effects **if the risk occurs**. Impact can be measured in terms of:

- **Cost** (e.g., \$20,000 over budget)
- **Time** (e.g., 2-week delay)
- **Scope/Quality** (e.g., reduced features, more defects)
- **Reputation, safety**, etc.

SCALE	PROBABILITY	+/- IMPACT ON PROJECT OBJECTIVES		
		TIME	COST	QUALITY
Very High	>70%	>6 months	>\$5M	Very significant impact on overall functionality
High	51-70%	3-6 months	\$1M-\$5M	Significant impact on overall functionality
Medium	31-50%	1-3 months	\$501K-\$1M	Some impact in key functional areas
Low	11-30%	1-4 weeks	\$100K-\$500K	Minor impact on overall functionality
Very Low	1-10%	1 week	<\$100K	Minor impact on secondary functions
Nil	<1%	No change	No change	No change in functionality

Probability and impact matrix

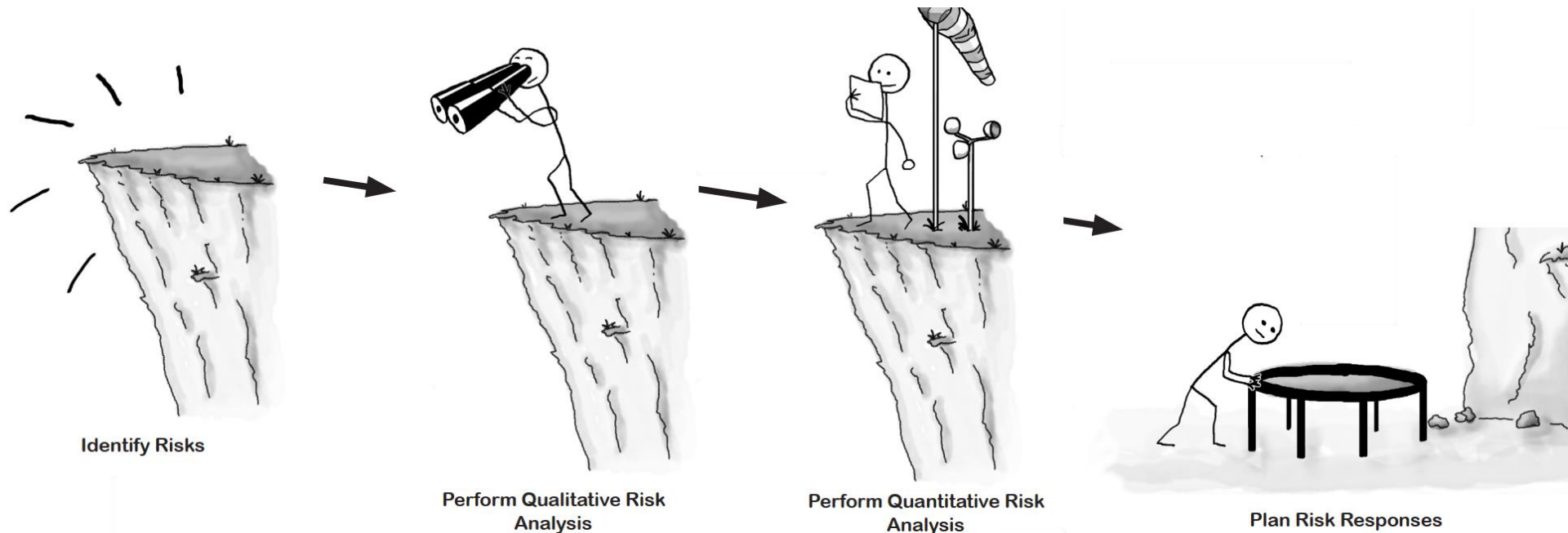
		Threats					Opportunities						
Probability	Very High 0.90	0.05	0.09	0.18	0.36	0.72	0.72	0.36	0.18	0.09	0.05	Very High 0.90	Probability
	High 0.70	0.04	0.07	0.14	0.28	0.56	0.56	0.28	0.14	0.07	0.04	High 0.70	
	Medium 0.50	0.03	0.05	0.10	0.20	0.40	0.40	0.20	0.10	0.05	0.03	Medium 0.50	
	Low 0.30	0.02	0.03	0.06	0.12	0.24	0.24	0.12	0.06	0.03	0.02	Low 0.30	
	Very Low 0.10	0.01	0.01	0.02	0.04	0.08	0.08	0.04	0.02	0.01	0.01	Very Low 0.10	
		Very Low 0.05	Low 0.10	Moderate 0.20	High 0.40	Very High 0.80	Very High 0.80	High 0.40	Moderate 0.20	Low 0.10	Very Low 0.05		
Negative Impact						Positive Impact							

A Probability and Impact Matrix is a tool used in qualitative risk analysis to prioritize risks

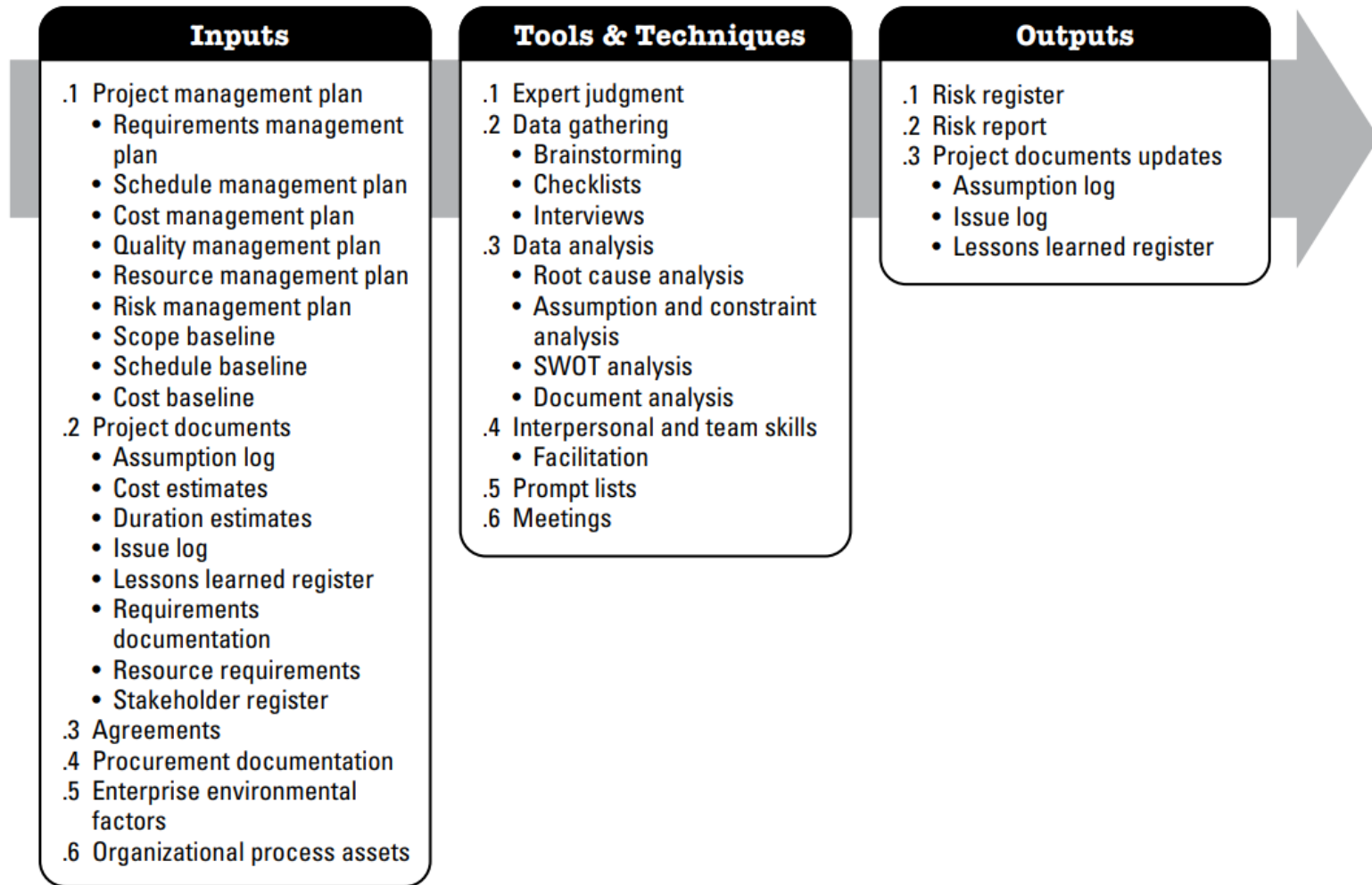
Identify Risk

Identify Risk

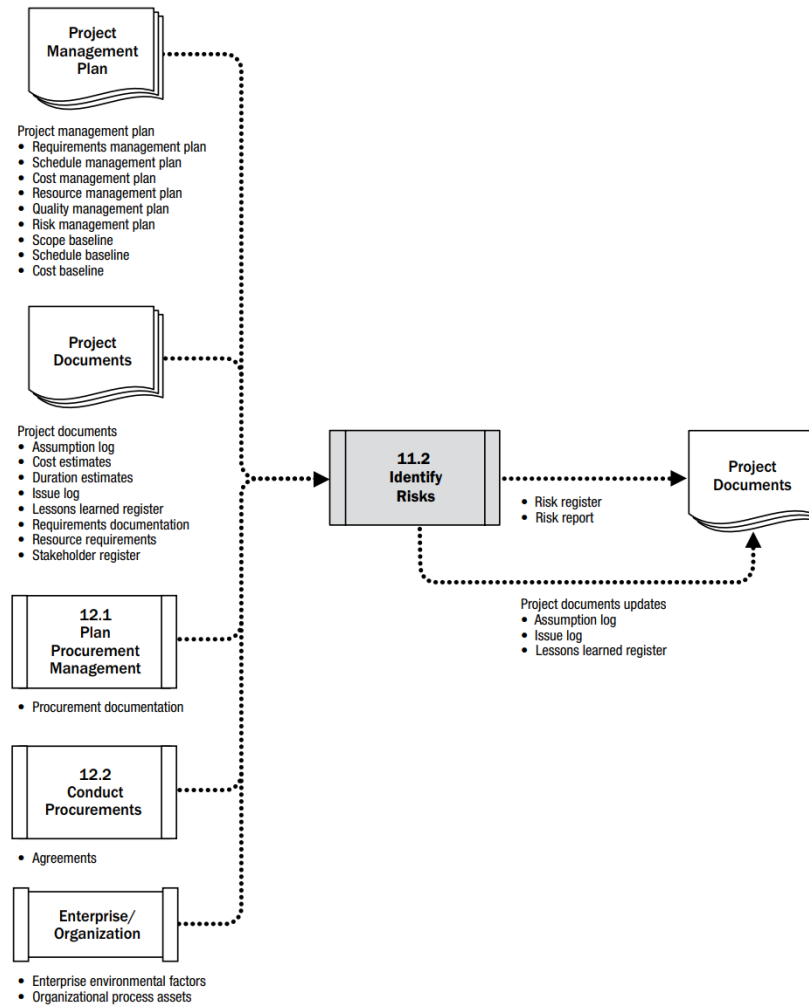
- **What:** process of identifying individual project risks as well as sources of overall project risk, and documenting their characteristics.
- **Why:** documentation of existing individual project risks and the sources of overall project risk.
- **When:** performed throughout the project.



Process



Data Flow



Tools & Techniques: Data Gathering

Brainstorming – facilitated workshops of multidisciplinary participants to generate and record potential risks.

Interview – Risks can be identified by interviewing project participants beyond the immediate team, other stakeholders and subject matter experts.



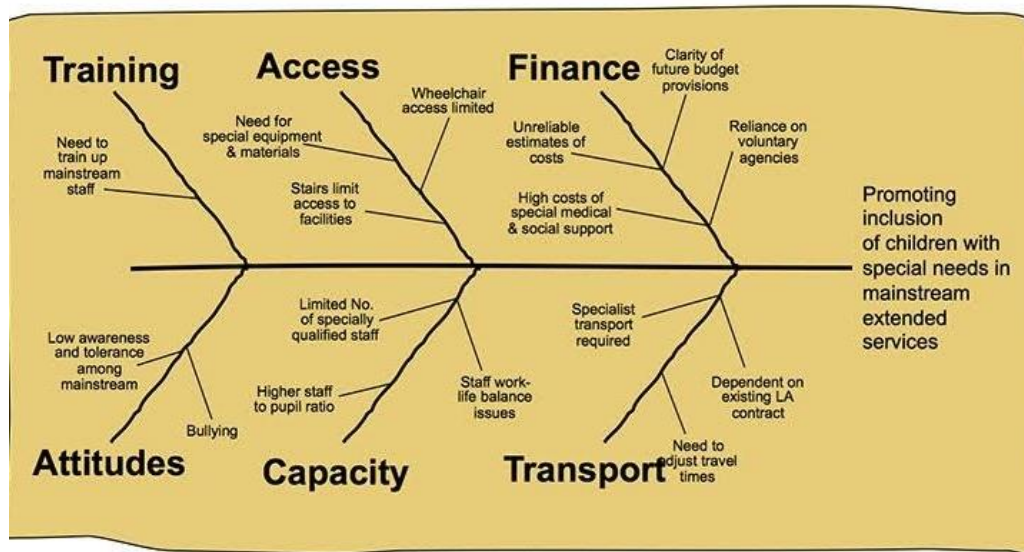
Checklists – Using lessons learned documents, old risk registers and issue logs from similar projects. Sometimes generic industry lists of common risks are available to help prompt ideas. The RBS is a good place to start for this.

Name of Event: _____ Date of Event: _____ Location: _____
Risk Management Checklist

Description	Risk Present?	Specific Risks	How will you limit or remove this risk?
Access to Facilities and Restrooms	☐		
Accommodation Requirements / ADA Compliance and Access	☐		
Alcohol Consumption	☐		
Cash Handling	☐		
Contracts Involved	☐		
Disruptive or Aggressive Individuals	☐		
Electrical/Lighting Concerns/Heating or Cooling	☐		
Excessive Noise or Amplified Sound (potential complaints)	☐		
Excessive Trash or Debris	☐		
First Aid / Emergency Response	☐		
Food Being Served/ Cooked	☐		

Tools & Techniques: Data Analysis

Root cause analysis with cause and effect diagrams – A technique designed to uncover the underlying risk. It often uses cause and effect diagrams such as fishbone diagrams.



Steps:

- 1. Define the Risk:** write the risk at the “head” of the diagram.
- 2. Identify Major Categories:** identify the main categories that contribute to the risk.
- 3. Brainstorm Contributing Factors:** write potential causes that contribute to the risk.
- 4. Analyze the Diagram:** identify key contributing factors.
- 5. Prioritize Risks:** assign priorities based on their likelihood and impact.
- 6. Develop Action Plans:** develop actionable plans to mitigate or eliminate risks.

Tools & Techniques: Data Analysis

Assumption and constraint analysis

– Project estimates of benefits and value delivered are based on assumptions and constraints. This activity examines these assumptions and constraints and asks what might jeopardize them or make them invalid. This can help identify risks.

Assumption: A factor in the planning process that is considered to be true, real, or certain, without proof or demonstration.

IF-THEN

Assumption & Constraint Testing

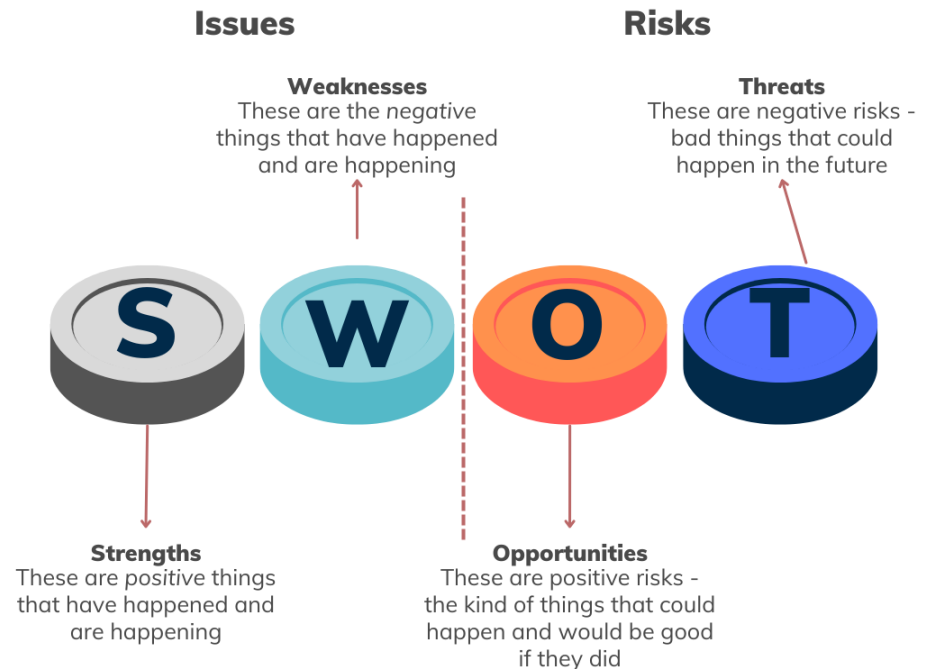
“IF this **assumption** proved to be false,
THEN the effect on the initiative/project would be ...”

“IF this **constraint** could be relaxed or removed,
THEN the effect on the initiative/project would be ...”



Tools & Techniques: Data Analysis

SWOT analysis – SWOT is an acronym for Strengths, Weaknesses, Opportunities and Threats. SWOT Analysis examines the project from each of these perspectives starting with the strengths and weaknesses of the project, organization and business area in general. From the strengths, opportunities are investigated, and ***the weaknesses examined as a potential source of threats.***



Tools & Techniques: Data Analysis

Document Analysis - Risks can be identified by reviewing project documents, such as plans, constraints, previous project files, contracts, agreements, and technical specifications.

Uncertainty, ambiguity or inconsistencies may be indicators of potential risks.



Tools & Techniques: Prompt list

Prompt lists

- A risk prompt list is a ***predefined checklist of risk categories*** used during risk identification in project management. It helps teams systematically explore different areas where risks might arise.
- The categories in the lowest level of the RBS or WBS can be used as a prompt list for individual project risks.
- Combine with techniques like brainstorming, interviews, and SWOT analysis

Prompt list categories

1. Technical
2. External
3. Organizational
4. Project Management
5. Legal/Compliance
6. Financial
7. Environmental/Social
8. Stakeholders



Category	Prompt Question	Example Risks
1. Technical	- Are we using new or unproven technologies? - Are system requirements unclear?	- Delay due to unfamiliar framework - Ambiguous requirements lead to rework
2. Requirements & Scope	- Is the scope clearly defined and agreed upon? - Are changes expected?	- Scope creep due to client change requests - Misunderstood user stories
3. Team & Resources	- Does the team have required skills? - Is key staff available throughout?	- Developer with key skills leaves project - Underestimation of resource needs
4. Schedule	- Is the timeline realistic? - Are milestones clearly defined?	- Tight deadlines cause rushed coding - Misaligned client and team schedules
5. Stakeholders	- Are stakeholders engaged and available? - Are their goals aligned?	- Conflicting priorities between business and IT - Stakeholder delay in feedback
6. Third-party Dependencies	- Are we relying on external vendors/APIs/tools? - Are SLAs clearly defined?	- API service goes down unexpectedly - Vendor delay affects our timeline
7. Quality & Testing	- Is there a test plan? - Is there enough time for QA and bug fixing?	- Insufficient test coverage - Defects discovered late in the release cycle
8. Security & Compliance	- Are there regulatory or security requirements? - Is data handling compliant?	- GDPR non-compliance - Security vulnerability due to poor input validation
9. Budget & Cost	- Is the budget fixed and approved? - Are there hidden or variable costs?	- Overtime cost overruns - Delays increase hosting costs
10. Tools & Infrastructure	- Are tools and environments ready? - Any risk with deployment or hosting?	- CI/CD pipeline failure - Server not provisioned in time

Tools & Techniques: Meeting

Risk workshop

- Most risk workshops include some form of brainstorming, but other risk identification techniques may be included
- Use of a skilled facilitator will increase the effectiveness of the meeting.



Tools & Techniques

Facilitation

A skilled facilitator can help participants remain focused on the risk identification task, follow the techniques accurately, ensure clear risk descriptions, identify and overcome sources of bias, and resolve any disagreements that may arise.

Facilitator

- Root word: *Facilitate*
- Origin: From Latin *facilis*, meaning "easy"
 - *Facilis* → "easy to do"
 - Related verb: *facere* → "to make or do"

So, to **facilitate** originally means “to make something easier”.

Thus, a **facilitator** is someone who **makes a process easier** for others.

Top 10 QUALITIES OF A GREAT FACILITATOR



Outputs: Risk Register

- The risk register may contain **limited** or **extensive** risk information depending on project variables such as size and complexity.
- On completion of the Identify Risks process, the content of the risk register may include but is not limited to:
 - ***Identified risks***
 - ***Potential risk owners***
 - ***Potential risk responses***
- Risk Register will get detailed and updated through other processes in Risk.

[illegible]

Outputs: Risk Report

The risk report presents information on sources of overall project risk, together with summary information on identified individual project risks.

May include but is not limited to:

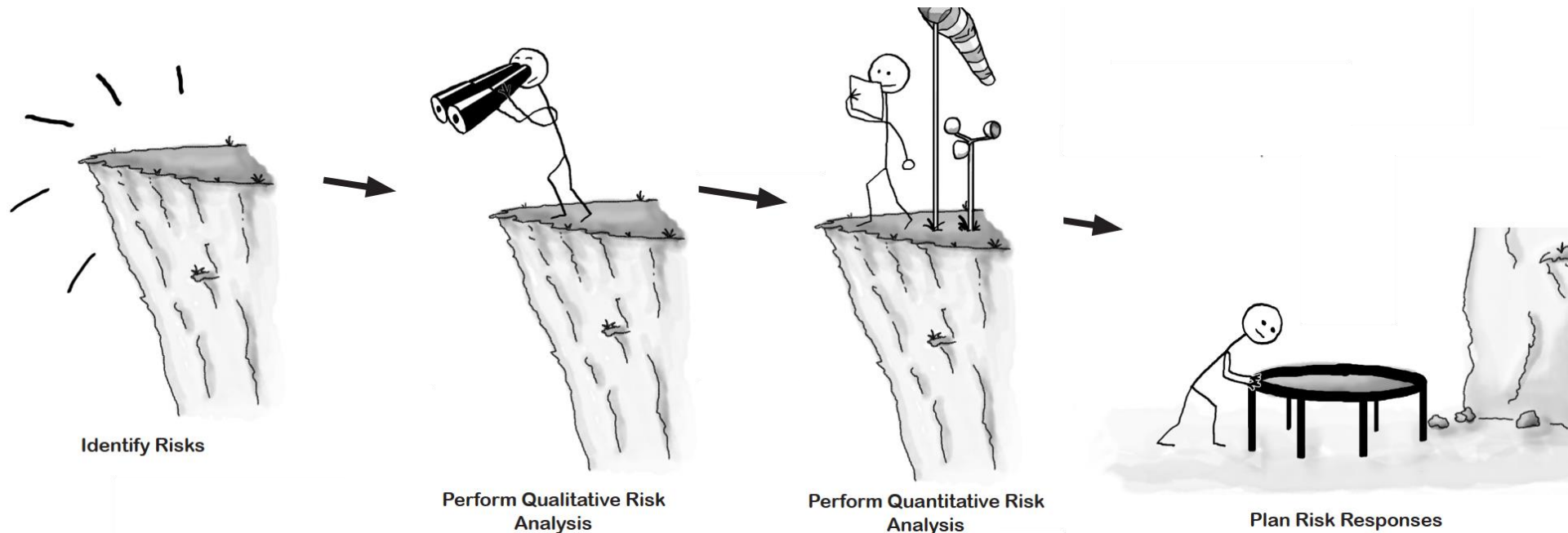
- **Sources of overall project risk**, indicating which are the most important drivers of overall project risk exposure; and
- Summary information on identified **individual project risks**, such as number of identified threats and opportunities, distribution of risks across risk categories, metrics and trends, etc.
- **Additional information** may be included in the risk report, depending on the reporting requirements specified in the risk management plan.

RISK REPORT					
Project Title: _____			Date: _____		
Executive Summary					
OVERALL PROJECT RISK					
Overall Risk Status and Trends					
Significant Drivers of Overall Risk			Recommended Responses		
INDIVIDUAL PROJECT RISKS					
Indicate the number of individual risks in each box below.					
VH					
H					
M					
L					
VL					
	VL	L	M	H	VH
Page 1 of 4					

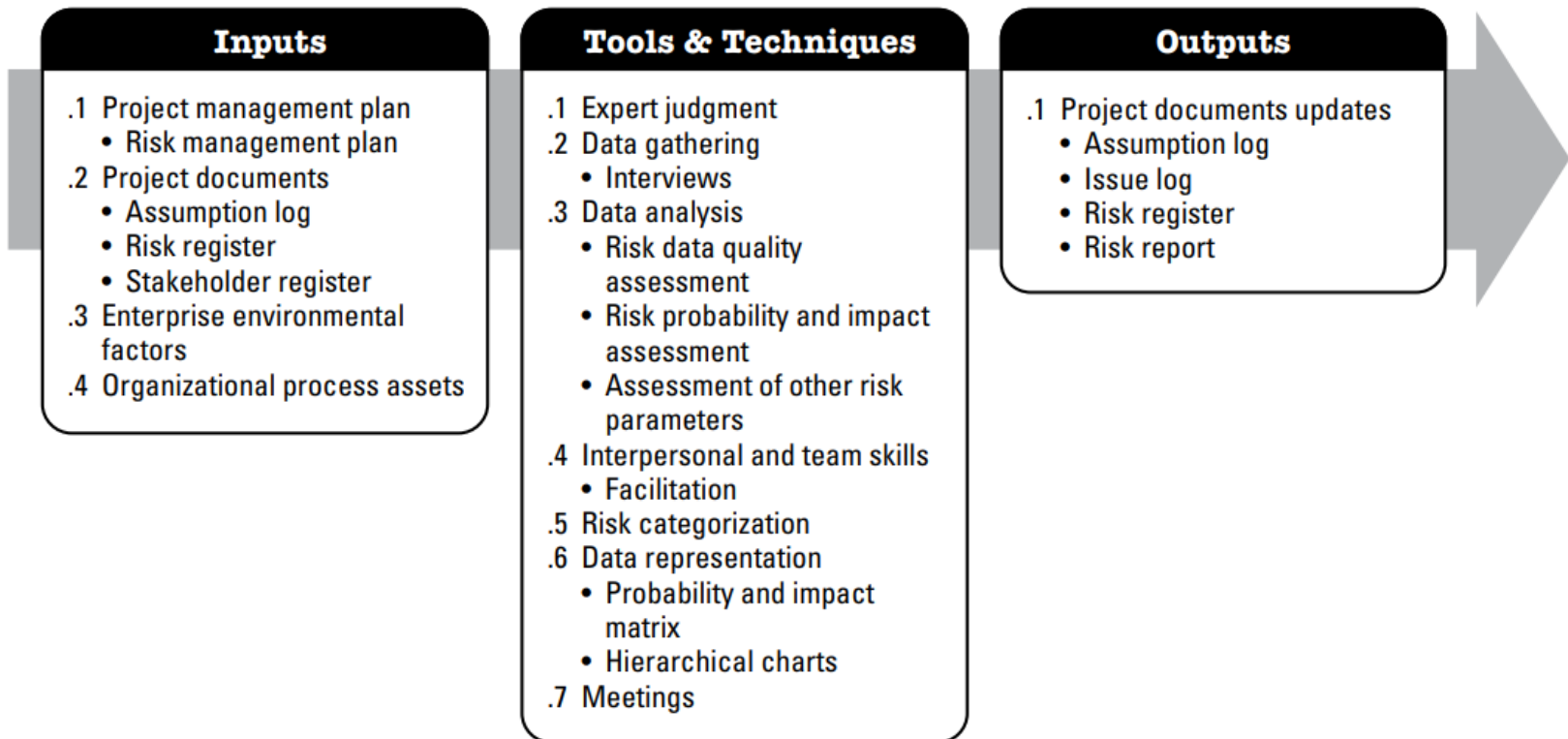
Perform Qualitative Risk Analysis

Perform Qualitative Risk Analysis

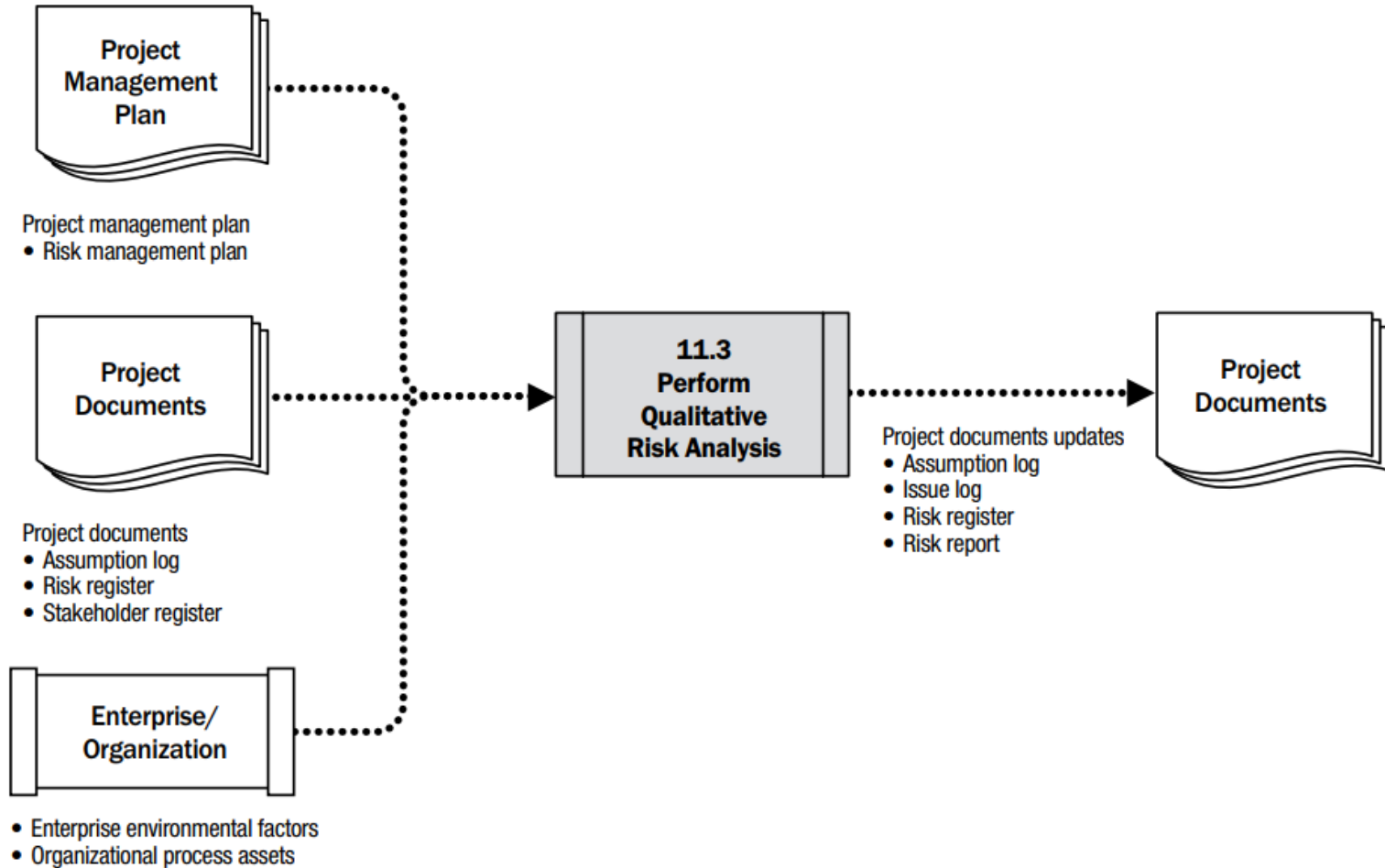
- **What:** process of prioritizing individual project risks for further analysis or action by assessing their probability of occurrence and impact as well as other characteristics.
- **Why:** focuses efforts on high-priority risks.
- **When:** performed throughout the project.



Process



Data Flow



Tools & Techniques: Data analysis

Risk data quality assessment

means making sure that the data you're using in your risk assessment is accurate. Sometimes it makes sense to bring in outside experts to check out the validity of your risk assessment data. Sometimes you can even confirm the quality of the data on your own, by checking some sample of it against other data sources



Step	Description	Example
1. Identify Risk Data Sources	Determine where the risk information comes from.	Interviews, past project data, expert judgment, vendor reports
2. Evaluate Data Completeness	Check if all necessary data is available.	Does the cost estimate include all modules? Are assumptions documented?
3. Evaluate Data Accuracy	Check if the data is correct and free from errors.	Are time estimates based on actual past performance or guesswork?
4. Evaluate Data Reliability	Assess how trustworthy and up-to-date the source is.	Is the source a recent expert study or a 5-year-old blog post?
5. Evaluate Data Consistency	See if the data aligns across sources.	Do risk ratings from different teams contradict each other?
6. Document Findings	Record data quality and gaps in a table or risk log.	Note that risk #5 is based on incomplete scope info.
7. Decide on Action	If data is weak, plan how to improve or validate it.	Schedule follow-up interview with client to clarify unclear requirement.

Tools & Techniques: Data analysis

Risk probability and impact assessment

- **Probability:** the likelihood that an event will occur.
- **Impact:** the amount of pain the risk event poses to the project.
 - Can be a relative scale (ex. high- medium-low) or a numeric scale known as a **cardinal scale**.
 - Organizations can give weightage for any specific parameter/s (Normally Scope, Cost, Quality and Time) in definition of score.

Scale	Probability	Probability score	Impact on Project			
			Schedule	Cost	Scope	Impact score
VLO	<10%	0.1	<2 weeks	<1%	Temporary defects, causing minor short term consequences	0.05
LO	10 to <30%	0.3	2 weeks to <1 month	1% to <2%	Product performance shortfall in area of tertiary (minor) importance	0.1
MED	30 to <50%	0.5	1 month to <2 months	2% to <4%	Product performance shortfall in area of secondary importance	0.2
HI	50 to <70%	0.7	2 months to <4 months	4% to <8%	Minor product performance shortfall in area of primary (critical) importance	0.4
VHI	70% plus	0.9	4 months plus	8% plus	Significant failure of product to meet one of its primary (critical) purposes	0.8

Tools & Techniques: Data analysis

Other risk parameters assessment

The project team may consider other characteristics of risk (in addition to probability and impact) when prioritizing individual project risks for further analysis and action.

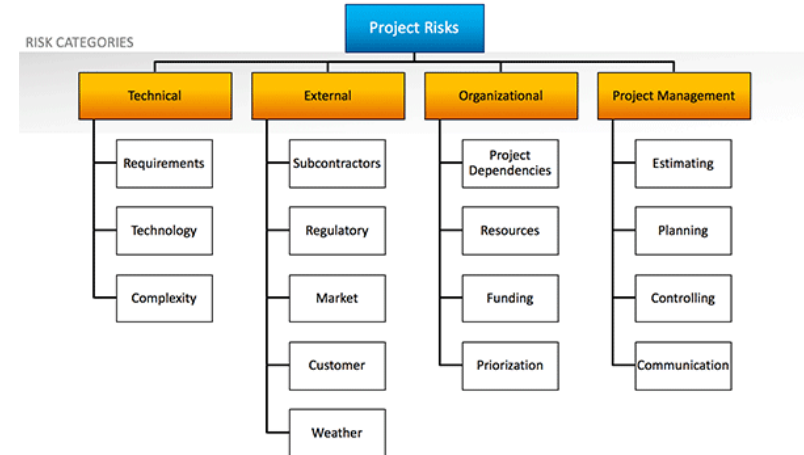
Risk Parameters	Meaning	Interpretations
Risk Urgency	Time period within which the risk is to be implemented.	Short time period = High urgency
Risk Proximity	Time period within which the risk will have an impact.	Short time period = High proximity
Risk Dormancy	Time period after the risk has occurred.	Short time period = Low Dormancy
Risk Manageability	Ease with which the risk can be managed.	Easy management = High manageability
Risk Controllability	Degree to which outcome of the risk can be controlled.	Easy control = High controllability
Risk Detectability	Ease with which the risk can be detected.	Easy detection = High detectability
Risk Connectivity	Extent to which a risk is connected to other risks.	More connection with other risks = High connectivity
Strategic Impact	Degree to which a risk can have impact on organization's strategic goal.	High effect/impact = High impact on strategy
Risk Propinquity	Degree to which the risk is perceived to matter by one or more stakeholders.	Significant for many stakeholders = High propinquity

Tools & Techniques: Data analysis

Risk Categorizations

- Grouping risks into categories can lead to the development of more effective risk responses by focusing attention and effort on the areas of highest risk exposure, or by developing generic risk responses to address groups of related risks.
- Identified risks can be categorized in many ways:
 - Project phases
 - Work Breakdown Structure(WBS)
 - Risk Breakdown Structure (RBS)
 - Common root causes

Risk Breakdown Structure (RiBS)



Tools & Techniques: Data representation

Probability and Impact Matrix

This matrix specifies combinations of probability and impact that allow individual project risks to be divided into priority groups

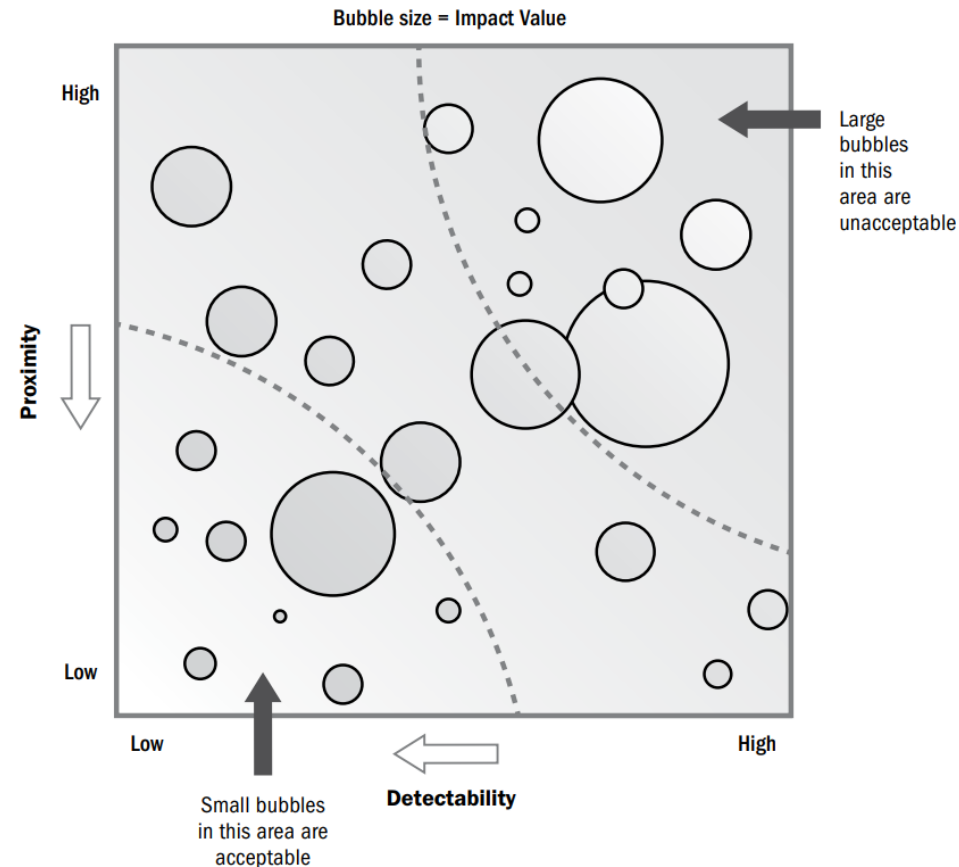
		Threats					Opportunities						
Probability	Very High 0.90	0.05	0.09	0.18	0.36	0.72	0.72	0.36	0.18	0.09	0.05	Very High 0.90	
	High 0.70	0.04	0.07	0.14	0.28	0.56	0.56	0.28	0.14	0.07	0.04	High 0.70	
	Medium 0.50	0.03	0.05	0.10	0.20	0.40	0.40	0.20	0.10	0.05	0.03	Medium 0.50	
	Low 0.30	0.02	0.03	0.06	0.12	0.24	0.24	0.12	0.06	0.03	0.02	Low 0.30	
	Very Low 0.10	0.01	0.01	0.02	0.04	0.08	0.08	0.04	0.02	0.01	0.01	Very Low 0.10	
Negative Impact						Positive Impact							

Tools & Techniques: Data representation

Hierarchical charts

Used when risk are categorized using more than two parameters.

Bubble chart: displays three dimensions of data, where each risk is plotted as a disk (bubble); bubble size is risk impact.



Outputs

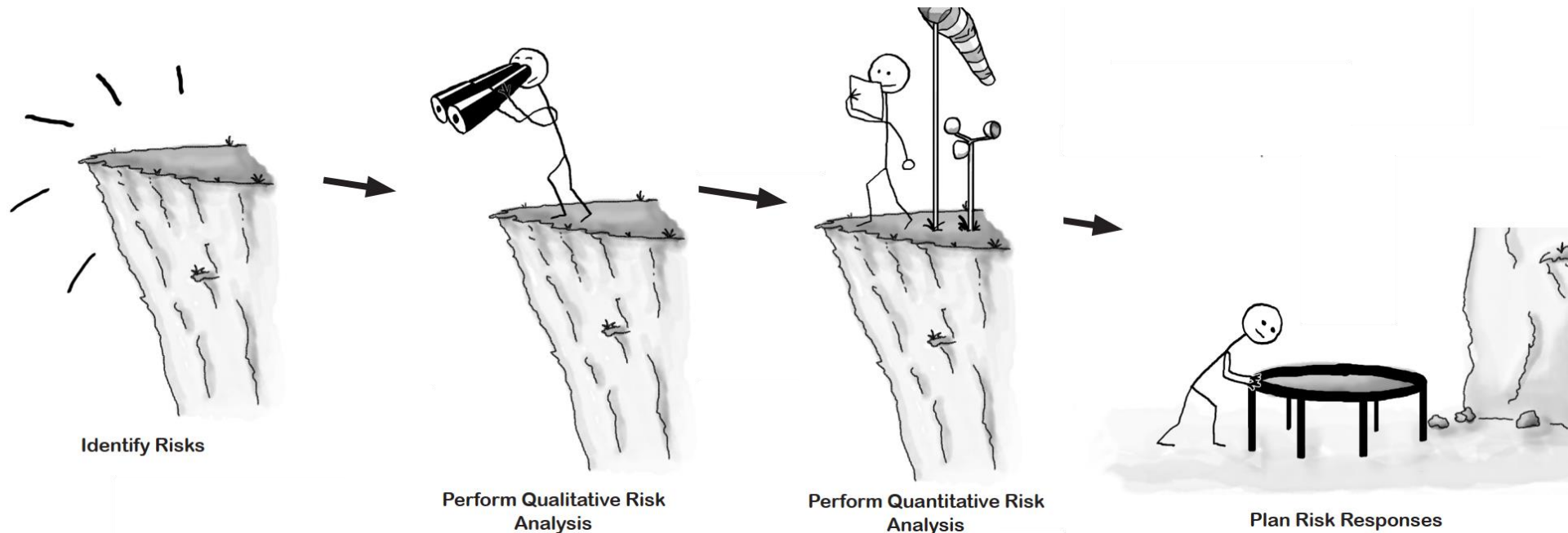
Project Document updates

- **Risk Register updates:** You may update the risk register with the following information:
 - Risks grouped by categories
 - Risk ranking (or priority/ risk score) for the identified risks
 - The nominated risk owner
 - List of risks requiring near-term responses
 - List of risks for additional analysis and response
 - Watch list of low-priority risks
 - Trends in Qualitative Risk Analysis results
- **Risk report.** The risk report is updated to reflect the most important individual project risks (usually those with the highest probability and impact), as well as a prioritized list of all identified risks on the project and a summary conclusion.
- **Assumption log.** new assumptions may be made, new constraints may be identified, and existing assumptions or constraints may be revisited and changed.
- **Issue log:** any new issues uncovered or changes in currently logged issues.

Perform Quantitative Risk
Analysis

Perform Quantitative Risk Analysis

- **What:** process of numerically analyzing the combined effect of identified individual project risks and other sources of uncertainty on overall project objectives.
- **Why:** It quantifies overall project risk exposure, and it can also provide additional quantitative risk information to support risk response planning.
- **When:** performed throughout the project (not required for all projects).



Perform Quantitative Risk Analysis

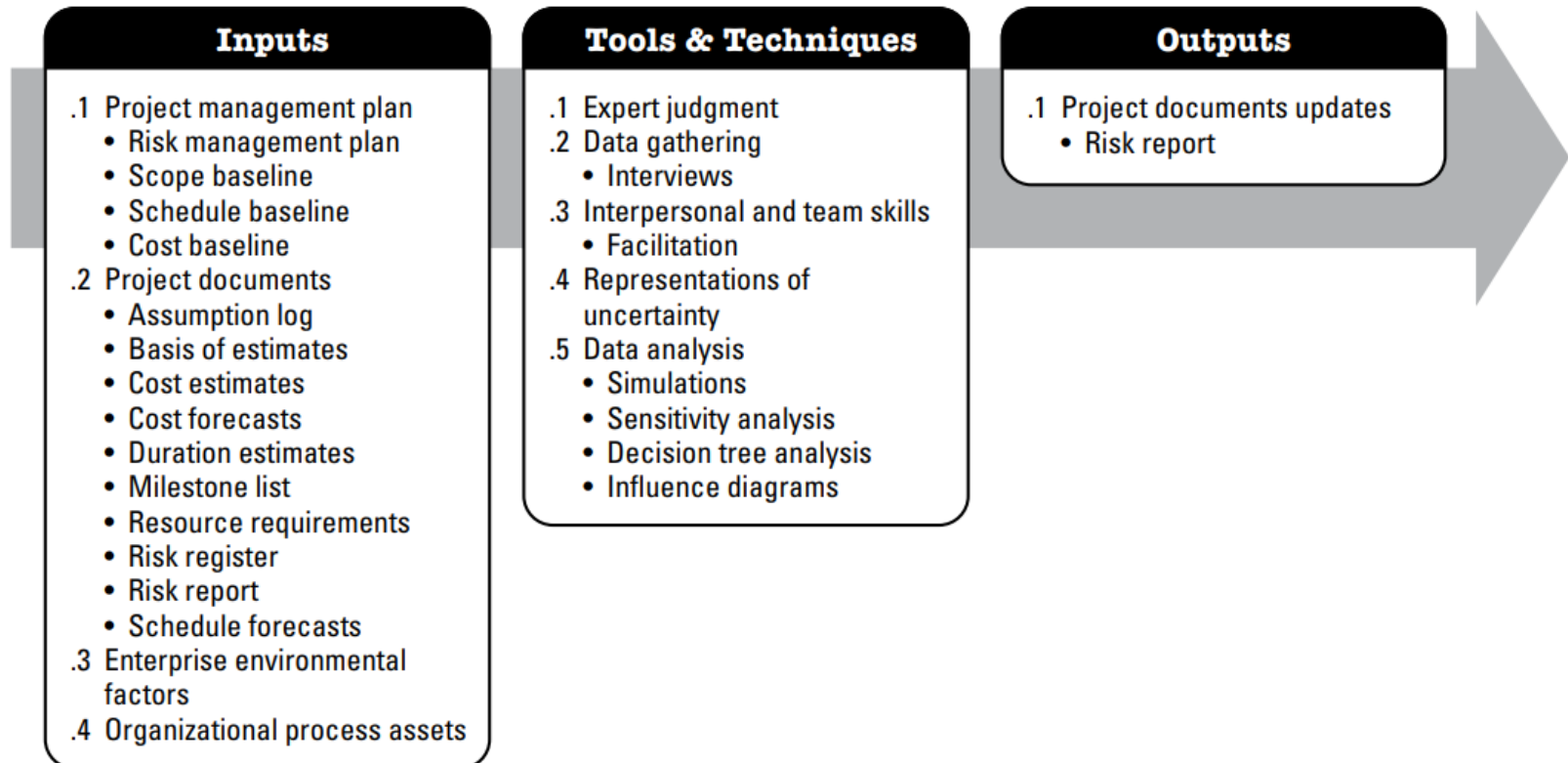
How?

- Translating information on individual project risks and other sources of uncertainty into **numeric inputs** for the quantitative risk analysis model
- Selecting the most appropriate **representation of uncertainty**
- Identifying which **tools** for the selected modeling techniques
- **Modeling the** risks or other sources of uncertainty in the context of the project
- **Interpreting** the outputs of quantitative risk analysis.

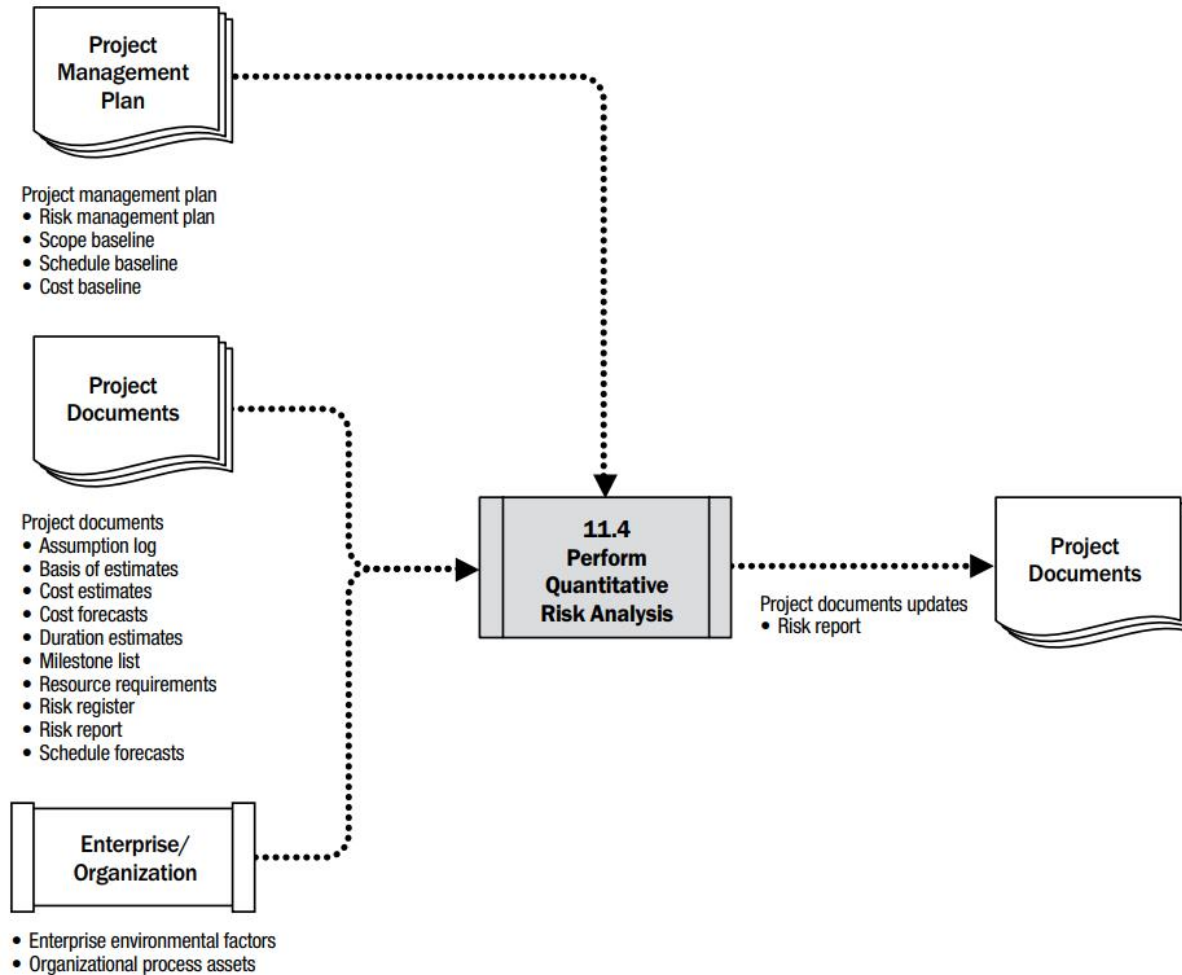


*Quantitative risk analysis is usually very lengthy and difficult and hence done for **only high priority risks.***

Process



Data Flow

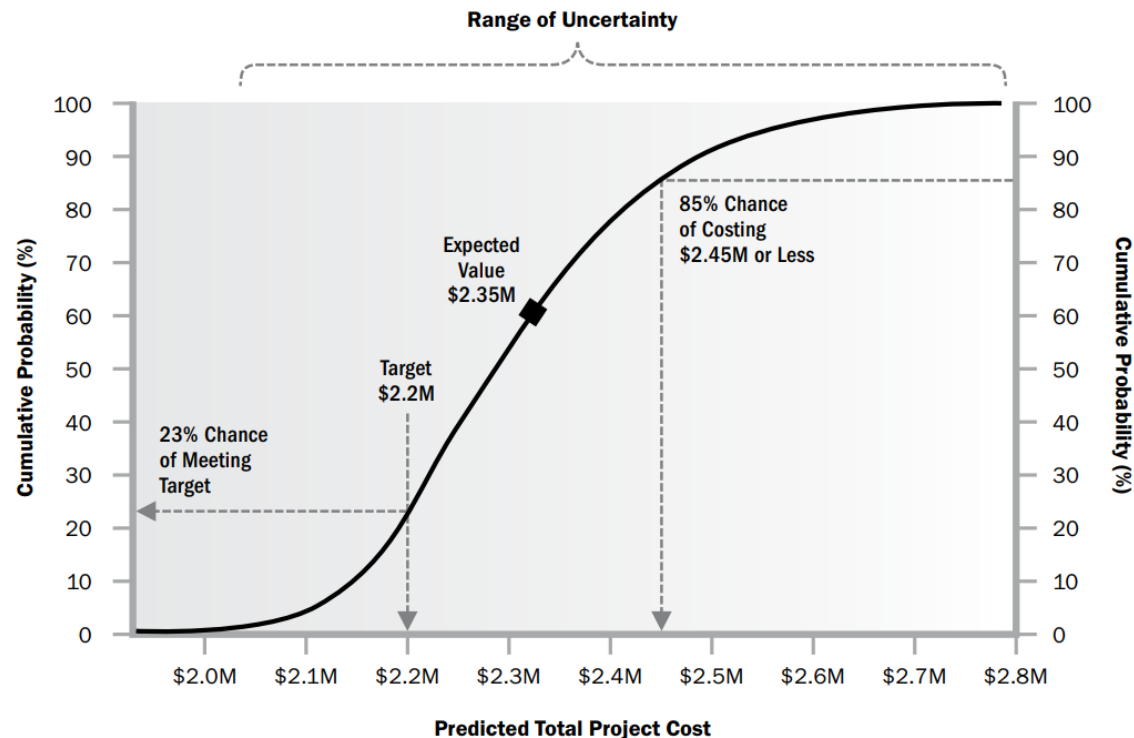


Tools & Techniques: Data analysis

Simulation

Quantitative risk analysis uses a model that simulates the combined effects of individual project risks and other sources of uncertainty to evaluate their potential impact on achieving project objectives.

Ex. Monte Carlo analysis for project cost

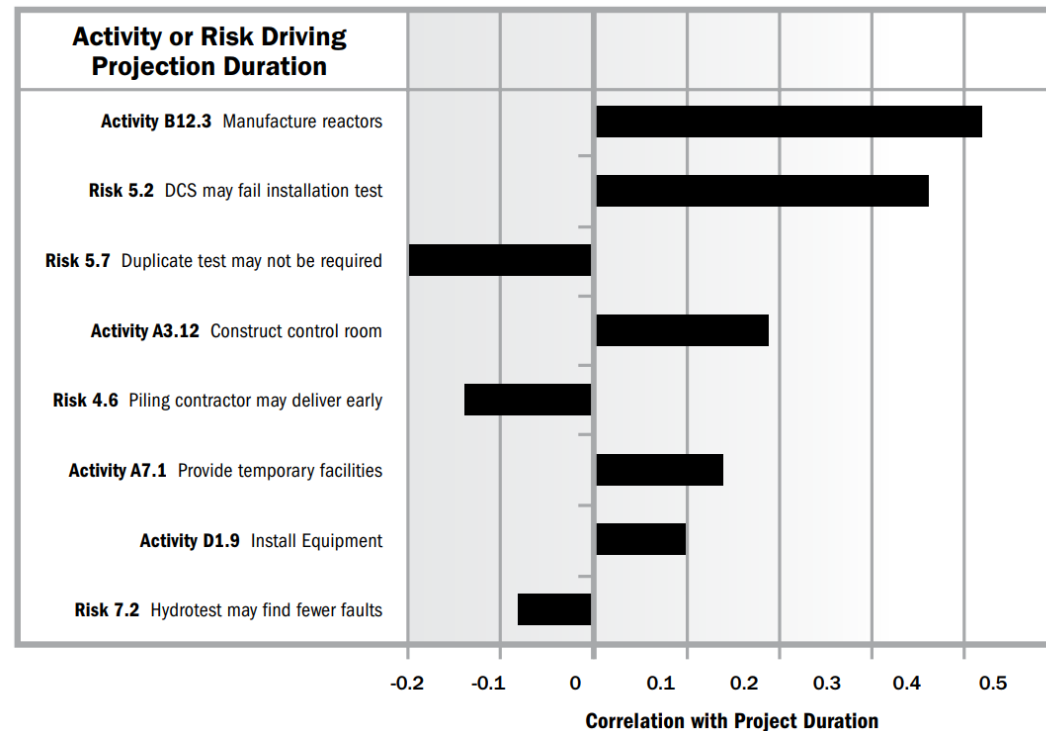


Tools & Techniques: Data analysis

Sensitivity analysis

A method of analyzing the potential impact of risk events on the project and determining *which risk has the greatest potential for impact*.

- All other risks are kept at the baseline while analyzing one risk
- Sensitivity analysis helps to determine which individual project risks or other sources of uncertainty have the **most potential impact** on project outcomes.

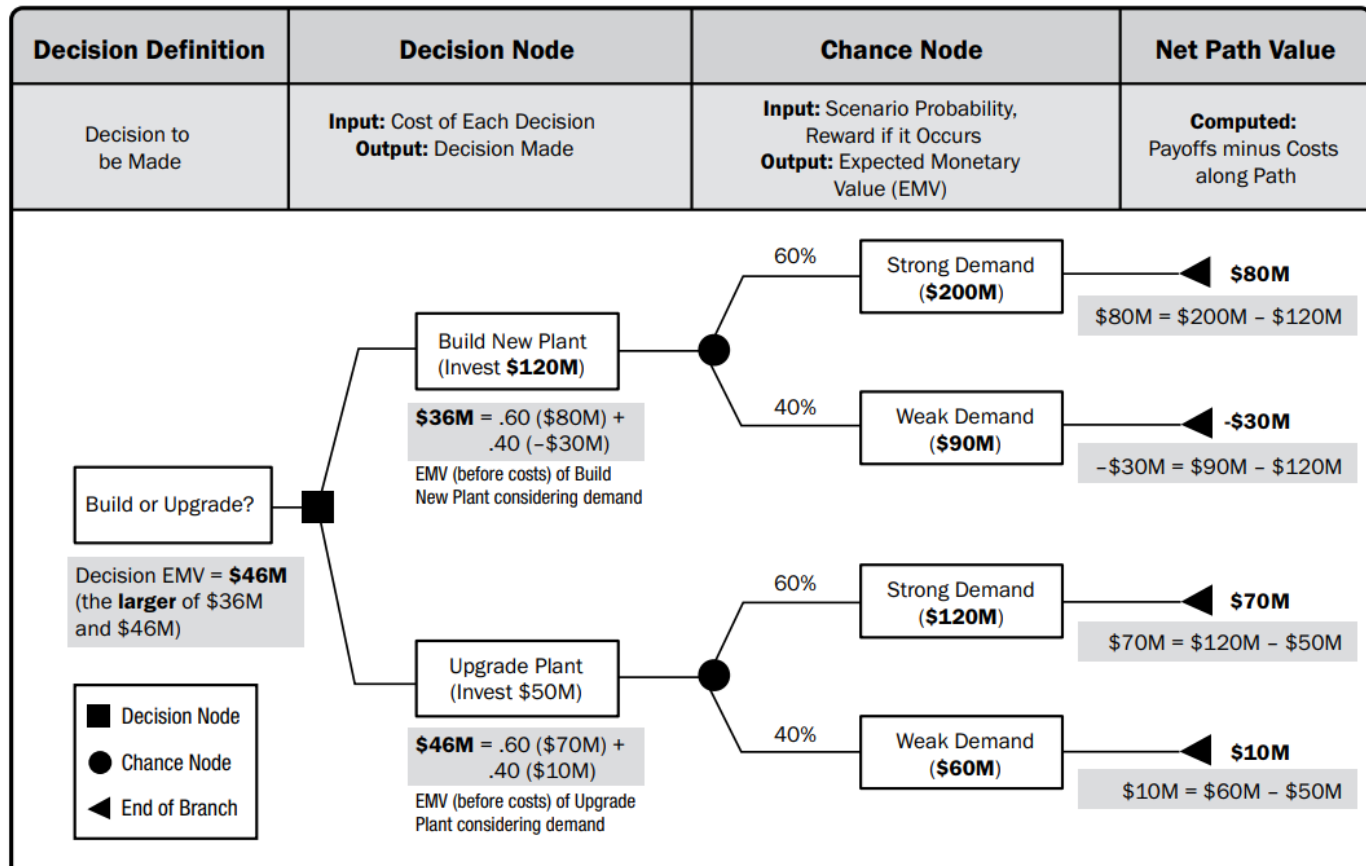


Tornado diagram

Tools & Techniques: Data analysis

Decision tree analysis

Decision trees are used to support selection of the best of several alternative courses of action.



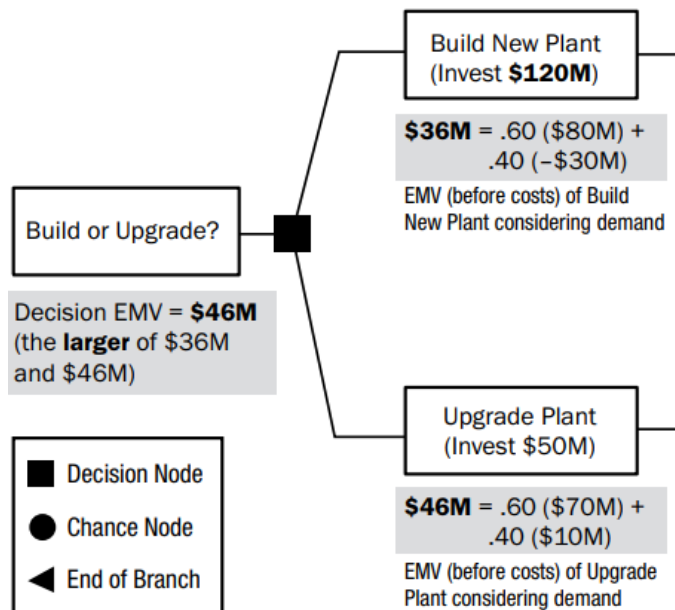
Net Path Value

Tools & Techniques: Data analysis

Expected Monetary Value (EMV)

A statistical technique in risk management that is used to quantify the risks, which in turn, assists the project manager to calculate the contingency reserve.

$$\text{EMV} = (\text{Probability} \times \text{Impact})$$



In previous example, from the calculations in the shaded areas, the upgraded plant has a higher EMV of \$46M – also the EMV of the overall decision. (This choice also represents the lowest risk, avoiding the worst case possible outcome of a loss of \$30M).

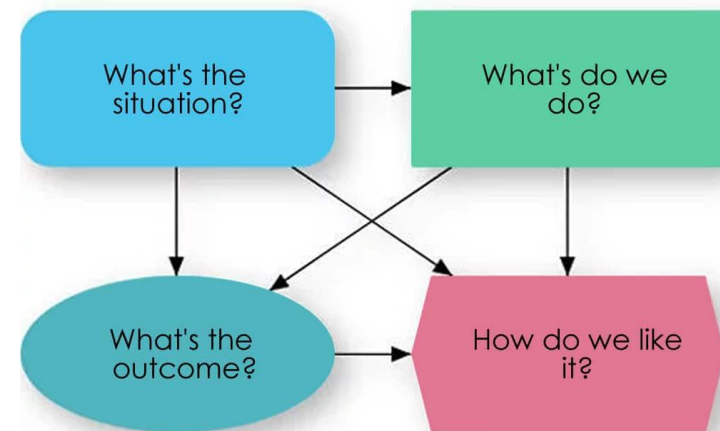
Tools & Techniques: Data analysis

Influence diagram

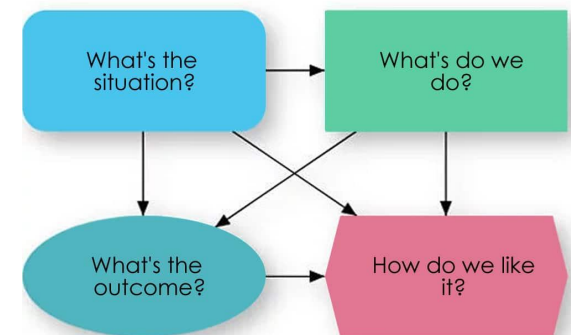
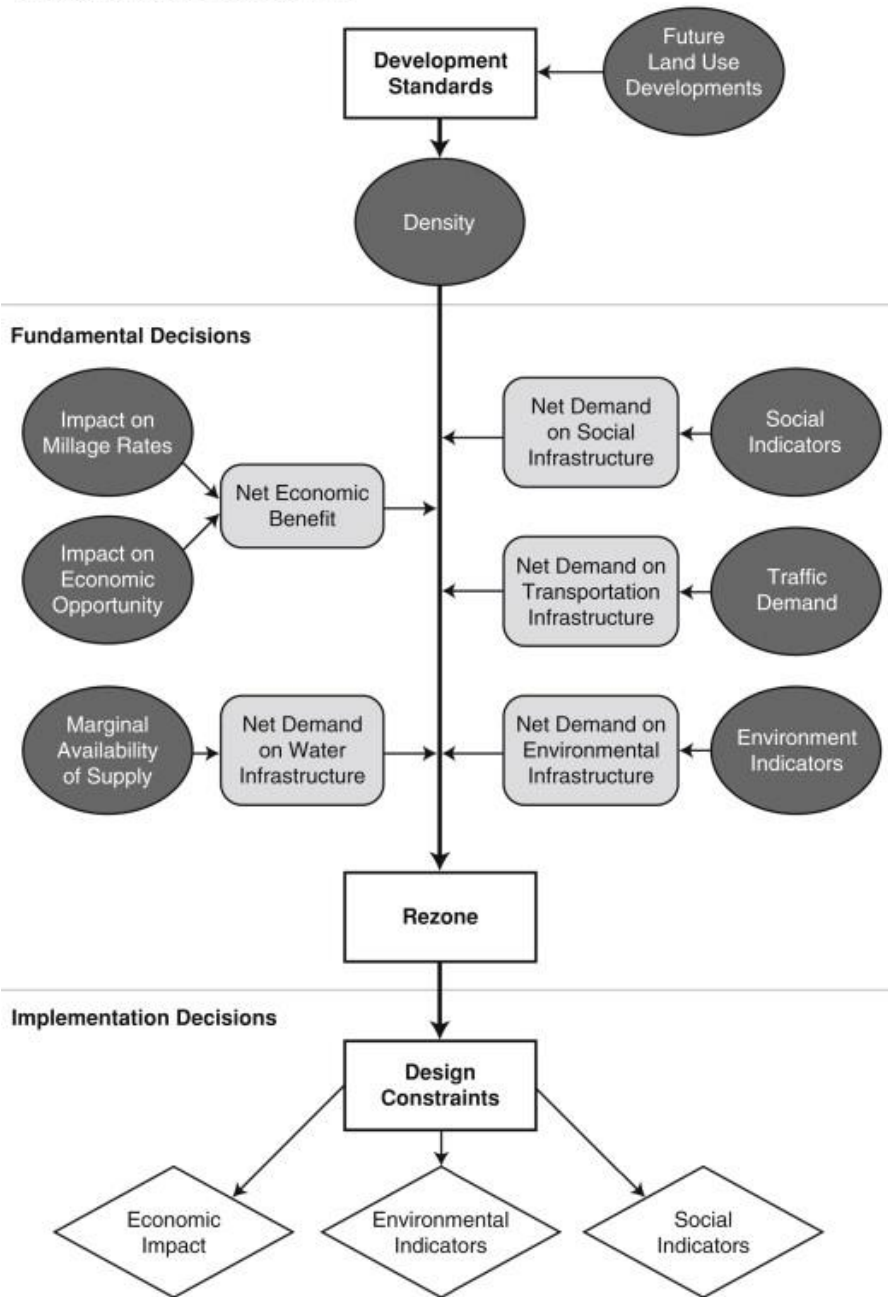
It depicts the key elements, including decisions, uncertainties, and objectives as nodes of various shapes and colors. It shows influences among them as arrows.

The influence diagram is then evaluated using a simulation technique, such as Monte Carlo analysis, to indicate which elements have the greatest influence on key outcomes.

- **Situation:** a *general variable* is a deterministic function of the quantities it depends on
- **Decision:** "What do we do?",
- **Chance variable:** "What's the outcome?",
- **Final valuation:** "How do we like it?"



Information Gathering Decisions

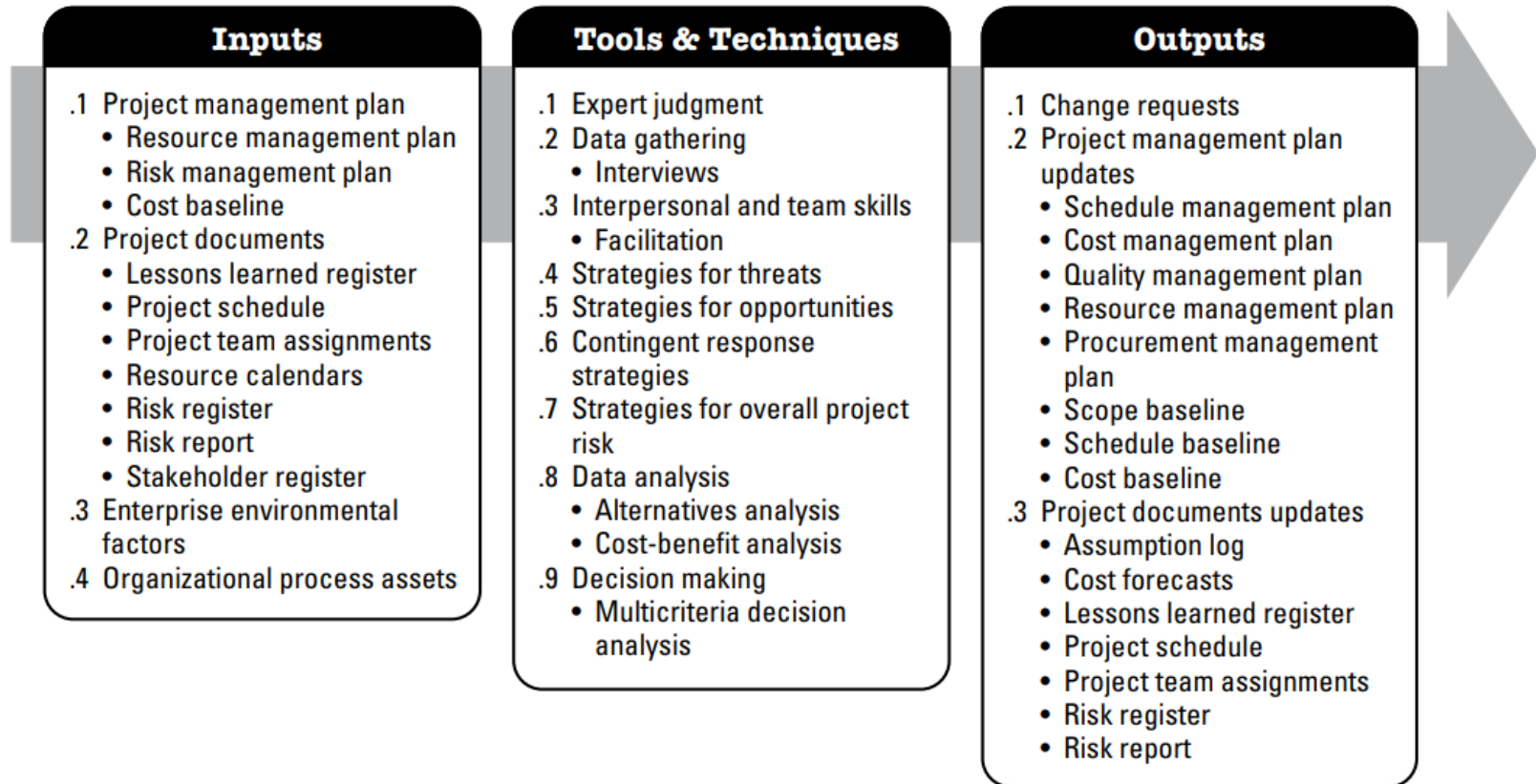


Plan Risk Responses

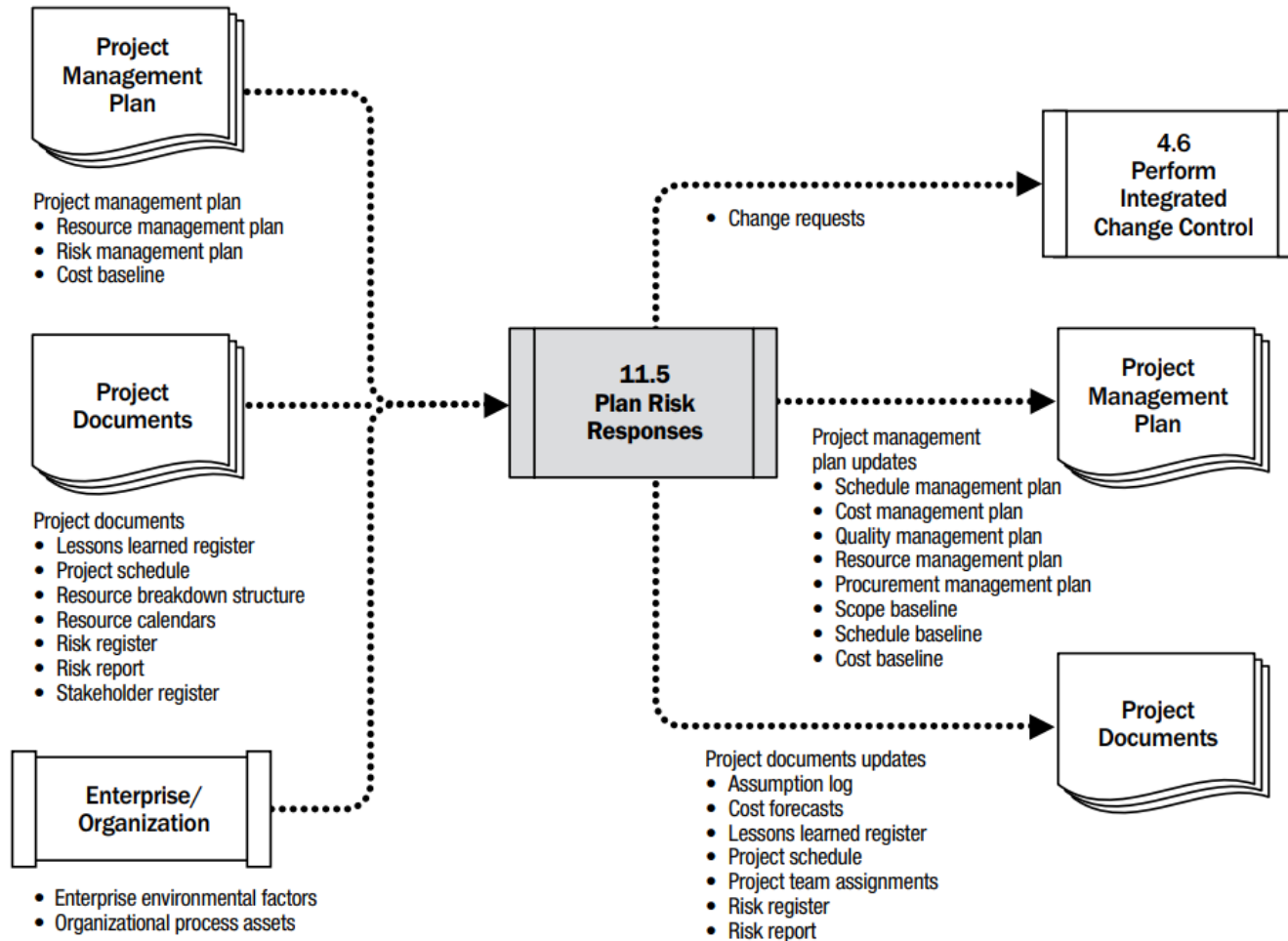
Plan Risk Responses

- **What:** process of developing options, selecting strategies, and agreeing on actions to address overall project risk exposure, as well as to treat individual project risks.
- **Why:** identifies appropriate ways to address overall project risk and individual project risks
 - Allocates resources and inserts activities into project documents and the project management plan as needed.
 - Effective and appropriate risk responses can minimize individual threats, maximize individual opportunities, and reduce overall project risk exposure.
- **When:** performed throughout the project
- **How:**
 - Risk responses should be appropriate for the significance of the risk, cost-effective in meeting the challenge, realistic within the project context, agreed upon by all parties involved, and owned by a responsible person.
 - Specific actions are developed to implement the agreed-upon risk response strategy, including primary and backup strategies, as necessary.

Process



Data Flow



Strategies for Threads

- **Escalate** – Transfer to someone outside the project. Convince the powers that be, that this risk is significant enough that it should be handled at the program or portfolio level.
- **Avoid** – take evasive action. Find a way to eliminate the source of the risk. Maybe use a different approach or technology that does not contain the risk.
- **Transfer** - Move the risk to a group better equipped to handle it. This can include taking out insurance which exchanges exposure for a known premium.
- **Mitigate**- Make the risk smaller. Through actions, reduce the probability of the risk occurring or the impact to the project should the risk occur.
- **Accept** – For low priority risks, or those we cannot avoid or reduce, we may have to accept the risk. Here the best thing we can do might be to create a contingency reserve to help deal with the risk should it occur



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Strategies for Opportunities

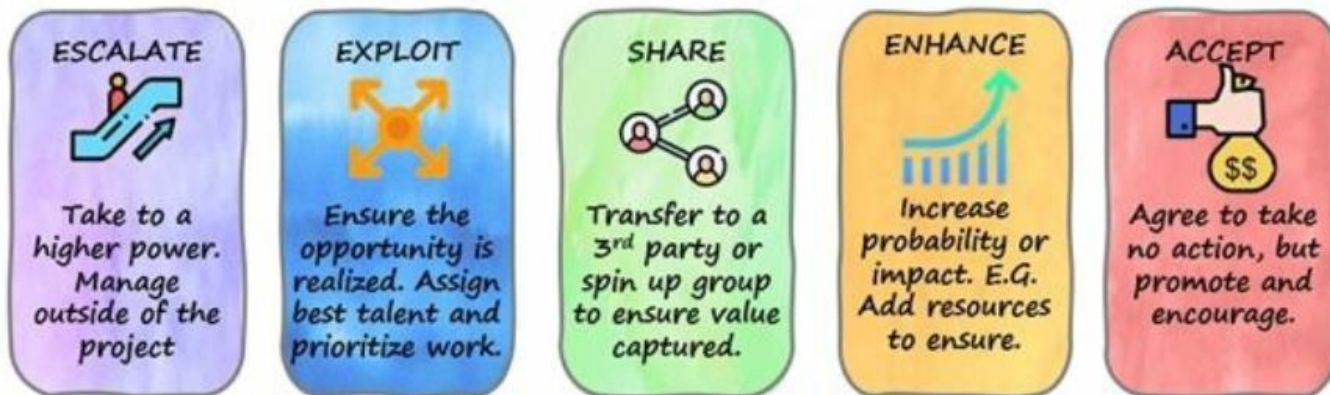
Escalate – Transfer to someone better able to ensure the opportunity occurs. Maybe someone with more influence or authority at the program, portfolio and sponsor level.

Exploit – Try to drive the probability up to 100% and maximize impact. Perhaps by assigning our best people to it early and giving them everything they need to ensure it occurs.

Share – Transfer the opportunity to others, either inside our organization or outside to maximize the value derived and captured.

Enhance – Similar to exploit, but to a lesser degree, try to increase the probability and impact. Focus attention on the causes and prioritize this work for early delivery.

Accept – Sometimes, there is not much we can do to increase the likelihood of an opportunity occurring (such as good weather). All we can do is be ready and prepared to take advantage of it, so it does not pass us by unrealized.



If the weather's good, then there's a chance you could see a meteor shower. If the team gets a photo that wins the meteor photo contest, you can get extra funding. You have your team stay up all night with their telescopes and cameras ready.

You hear that it's going to rain for the first three days of your trip, so you bring waterproof tents and indoor projects for the team to work on in the meantime.

You read that there's a major bear problem in the spring on the cliff where you are planning to work. You change your project start date to happen in the fall.

On your way up the cliff, you meet another team that is looking to survey the area. You offer to do half of the surveying work while they do the other half and then trade your findings with each other.

There's a high probability of water damage to some of your equipment, so you buy insurance to avoid losses.

There's always the chance that someone could make a mistake and fall off the cliff. No matter how much you plan for the unexpected, sometimes mistakes happen.

About 10 years ago a really rare bird, the black-throated blue warbler, was seen on this cliff. If you could get a picture of it, it would be worth a lot of money. So, you bring special seeds that you have read are really attractive to this bird, and you set up lookout points around the cliff with cameras ready to get the shot.

Mitigate

If the weather's good, then there's a chance you could see a meteor shower. If the team gets a photo that wins the meteor photo contest, you can get extra funding. You have your team stay up all night with their telescopes and cameras ready.

Avoid

You hear that it's going to rain for the first three days of your trip, so you bring waterproof tents and indoor projects for the team to work on in the meantime.

Accept

You read that there's a major bear problem in the spring on the cliff where you are planning to work. You change your project start date to happen in the fall.

Transfer

On your way up the cliff, you meet another team that is looking to survey the area. You offer to do half of the surveying work while they do the other half and then trade your findings with each other.

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Outputs

Project Management Plan Updates

- Contingency plans, fallback plans

Project Document Updates

- **Risk register updates:**
 - Response strategies
 - Residual risks
 - Secondary risks
 - The budget and schedule for risk responses
 - Risk owners and their assigned responsibilities

Project document updates

- **Risk related contract decisions**
 - The contract may be needed for insurance purposes, customer acceptance, or the acknowledgement of responsibilities between the entities completing the project.
 - More on procurement section
- **Assumptions log updates**

Change request

- Preventive actions

Contingent response Strategies

- Contingent response plan will be executed under certain predefined conditions:
Symptoms, warning signs or trigger.
- Events that trigger the contingency response, such as missing intermediate milestones or gaining higher priority with a supplier, should be defined and tracked.



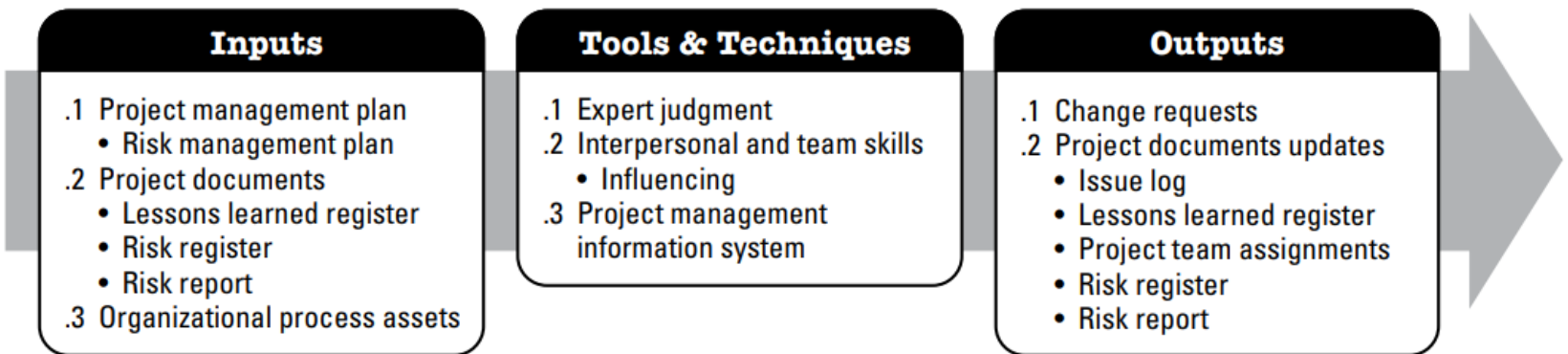
Risk Mitigation Plan	Risk Contingency Plan
You identify actions which you will take in advance irrespective of the occurrence of risk	You plan actions, but you monitor certain warning signs. You take these actions only when you see the warning signs.
You spend time and money in advance for the given risk condition	You do not spend time or money in advance, but you keep them ready and invest them when needed
We are expected to mitigate the risks which are outside the risk threshold. By applying a mitigation plan, we reduce the probability of the impact of the identified risk.	By identifying the contingency plan, we do not change the probability or impact of the current risk, but we plan to control the impact as risk event looks like occurring.
This works as the first level of defense for the high exposure risks	This works as a fallback plan for high exposure risks.

Implement Risk Responses

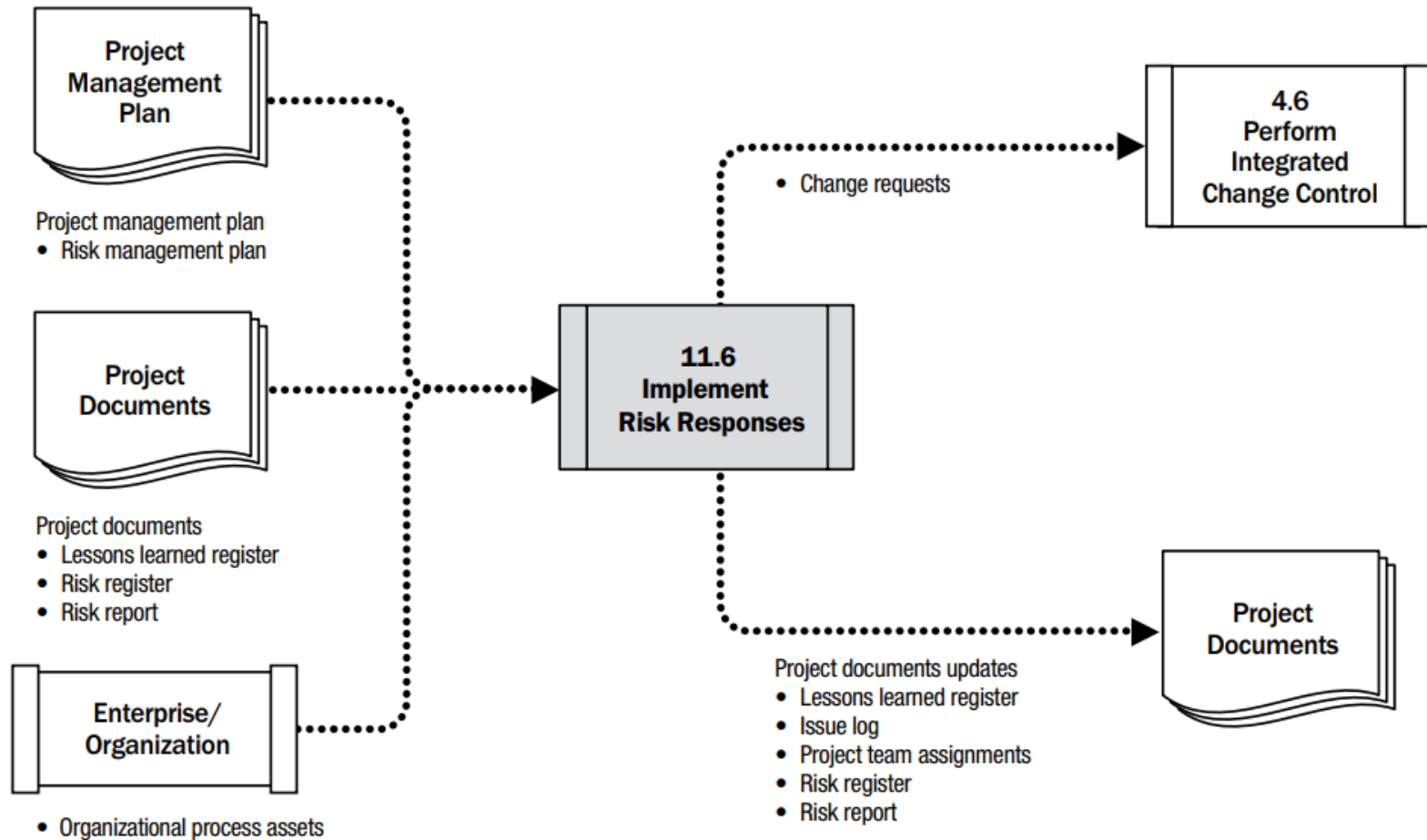
Implement Risk Responses

- **What:** process of implementing agreed-upon risk response plans.
 - A common problem with Project Risk Management is that project teams spend effort in identifying and analyzing risks and developing risk responses, then risk responses are agreed upon and documented in the risk register and risk report, but ***no action is taken to manage the risk.***
 - Only if risk owners give the required level of effort to implementing the agreed-upon responses will the overall risk exposure of the project and individual threats and opportunities be managed proactively.
- **Why:** ensures that agreed-upon risk responses are executed as planned in order to
 - address overall project risk exposure.
 - minimize individual project threats, and maximize individual project opportunities
- **When:** performed throughout the project

Process



Data Flow



Tools & Techniques

Influencing

The project manager or person responsible for facilitating the risk process may need to exercise influencing to encourage nominated **risk owners** to take necessary action where required.



Expert judgment

Expertise should be considered from individuals or groups with specialized knowledge to validate or modify risk responses if necessary, and decide how to implement them in the most efficient and effective manner.



Tools & Techniques

Project management information system (PMIS)

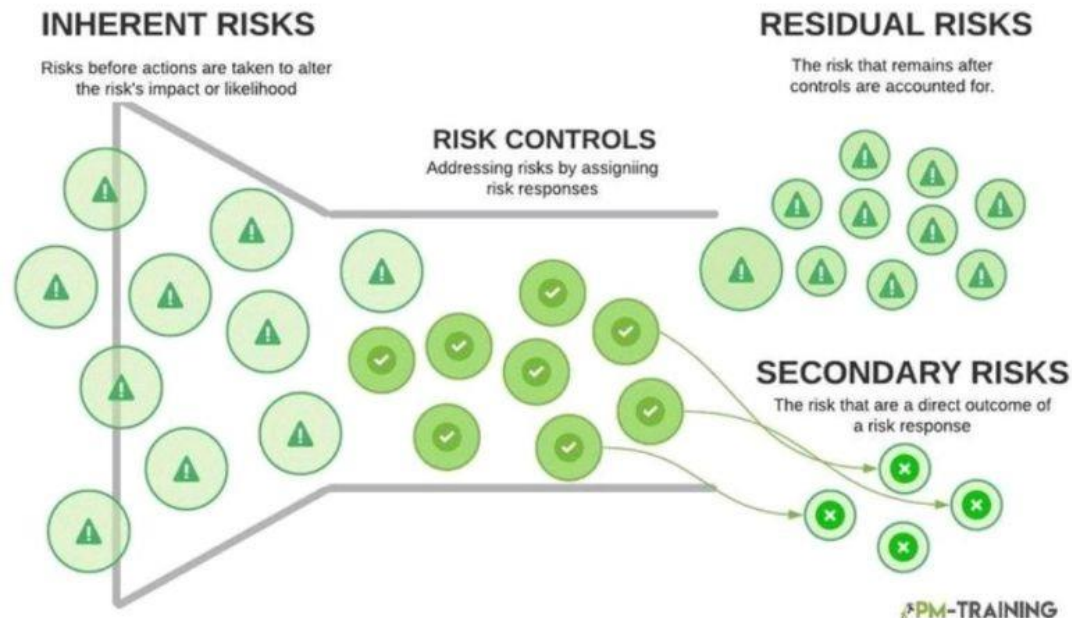
Schedule, resource, cost or risk management softwares may help to ensure that agreed-upon risk response plans and their associated activities are integrated into the project alongside other project activities.



Risk response can find even more risks

Secondary risks come from a response you have to another risk. If you dig a trench to stop landslides from taking out your camp, it's possible for someone to fall into the trench and get hurt

Residual risks remain after your risk responses have been implemented. So even though you reinforce your tent stakes and get weatherproof gear, there's still a chance that winds could destroy your camp if they are strong enough.



Outputs

Change requests

Change may happen to the cost and schedule baselines or other components of the project management plan.

Project documents updates

- **Risk register.** Any changes to the previously agreed-upon risk responses to any individual project risk
- **Risk report.** Any changes to the previously agreed-upon risk response to overall project risk exposure

Project documents updates

- **Issue log.**
- **Lessons learned register.**
- **Project team assignments.**
 - Suitably qualified and experienced personnel to execute the agreed-upon action (usually within the project team),
 - Any required technical resources to complete the action.

Monitor Risks

Monitor Risks

- **What:** process of monitoring the implementation of agreed-upon risk response plans, tracking identified risks, identifying and analyzing new risks, and evaluating risk process effectiveness throughout the project.
 - Even the best planning can't predict everything—there's always a chance that a new risk could crop up that you hadn't thought about.
- **Why:** enables project decisions to be based on current information about overall project risk exposure and individual project risks
 - Address overall project risk exposure.
 - Minimize individual project threats, and maximize individual project opportunities
- **When:** performed throughout the project
 - Risk monitoring should be done at **every status meeting**.

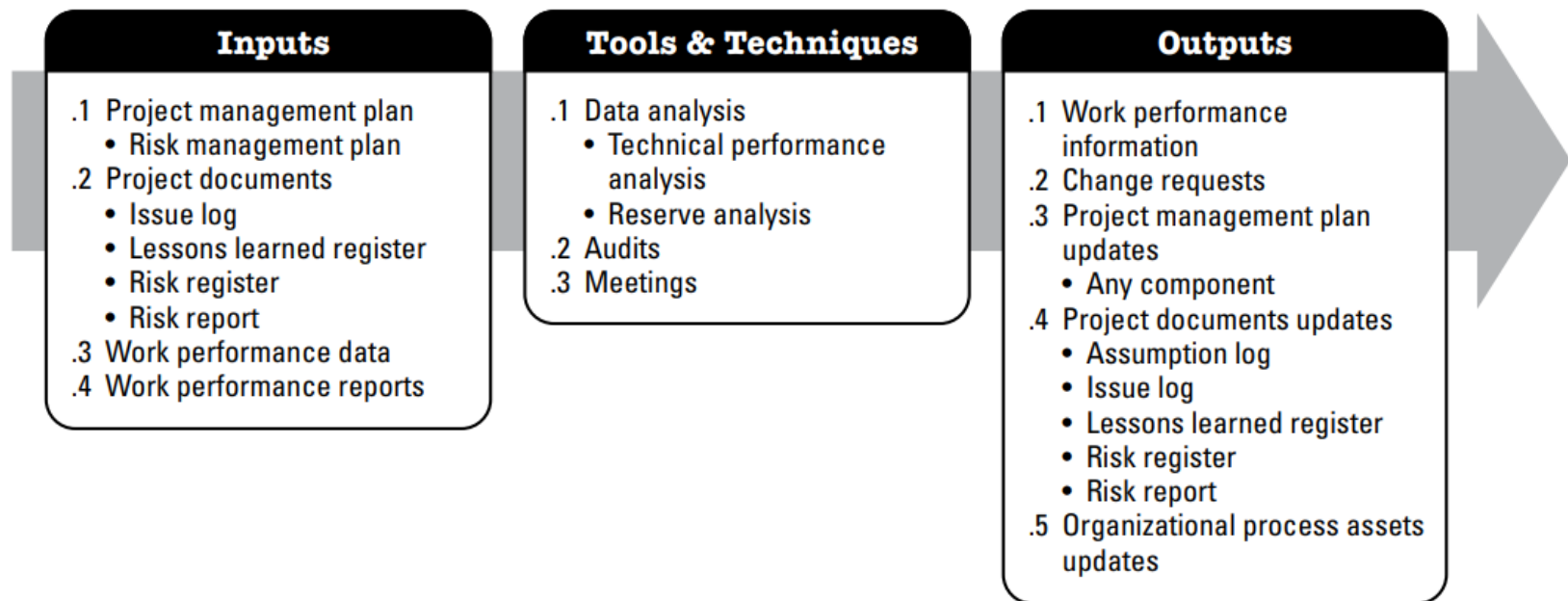


Monitor Risks

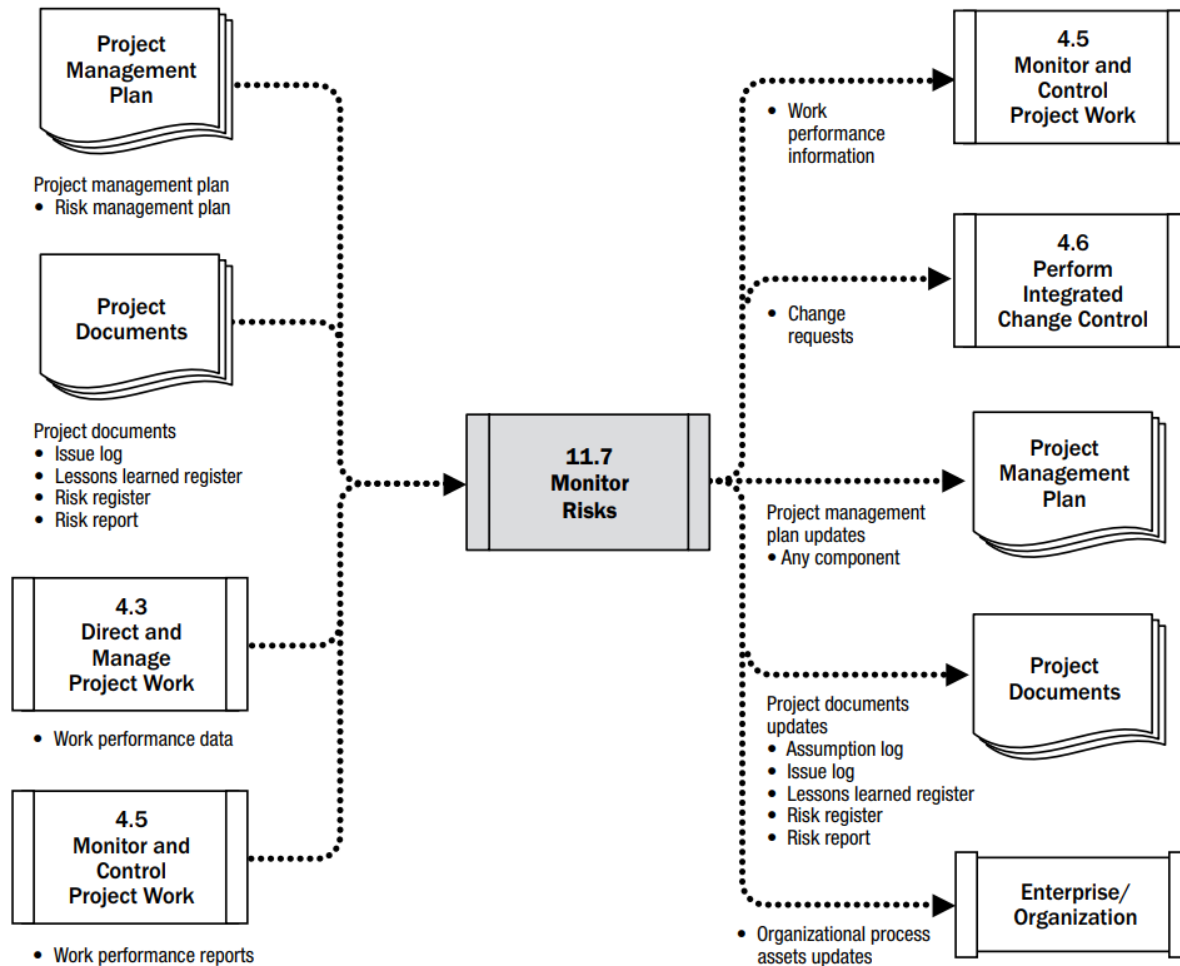
- **How:** The Monitor Risks process uses performance information generated during project execution to determine if:
 - Implemented risk responses are effective
 - Level of overall project risk has changed
 - Status of identified individual project risks has changed
 - New individual project risks have arisen
 - Risk management approach is still appropriate
 - Project assumptions are still valid
 - Risk management policies and procedures are being followed
 - Contingency reserves for cost or schedule require modification
 - Project strategy is still valid.



Process



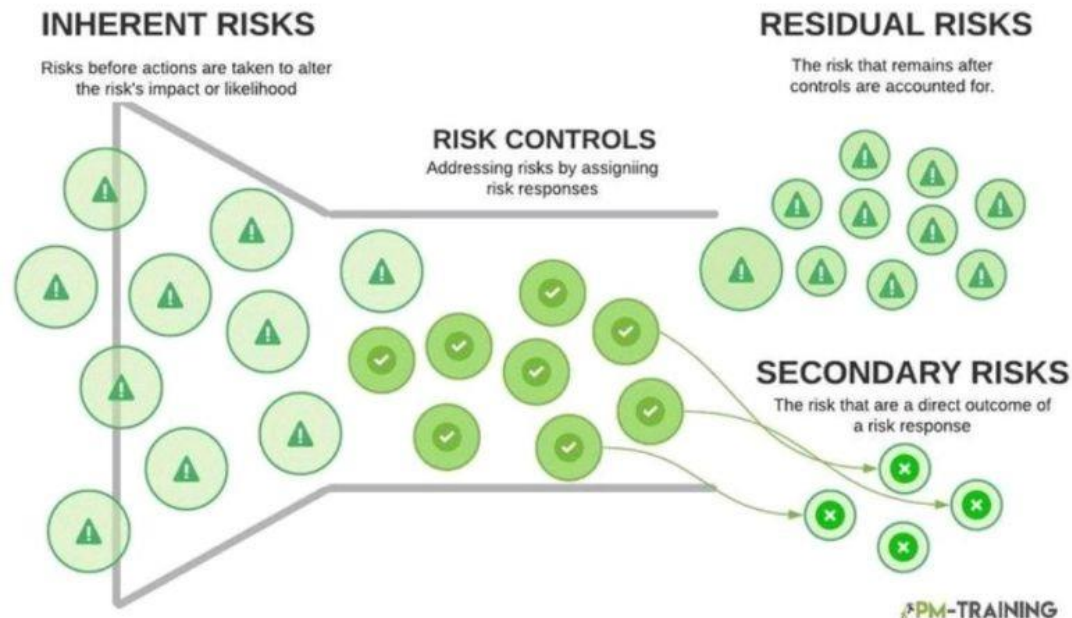
Data Flow



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Tools & Techniques

Meeting

- The periodic risk review is a regularly scheduled discussion throughout the project to
 - Identification of new risks,
 - Reassessment of current risks,
 - And the closing of risks that are outdated.
- The risk review may be conducted as part of a periodic project **status meeting** or a **dedicated risk review** meeting may be held, as specified in the risk management plan.

You should keep monitoring your risks at every meeting until the project is closed.

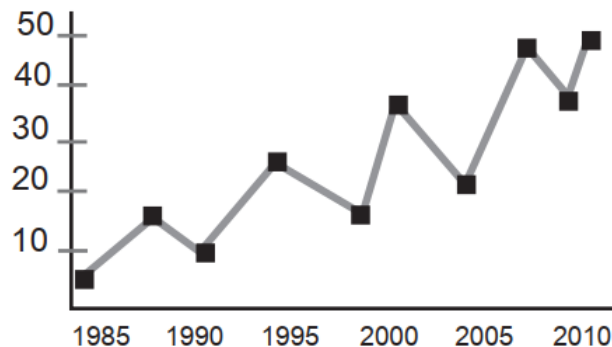
That's why the tools and techniques include status meetings.



Tools & Techniques

Technical performance analysis

- Technical performance measurement compares actual results against targets: weight, transaction times, number of delivered defects, storage capacity, etc
- Deviation can indicate the potential impact of threats or opportunities.
- Level of conformance can help to forecast the degree of success in achieving the project's scope.



Reserve Analysis

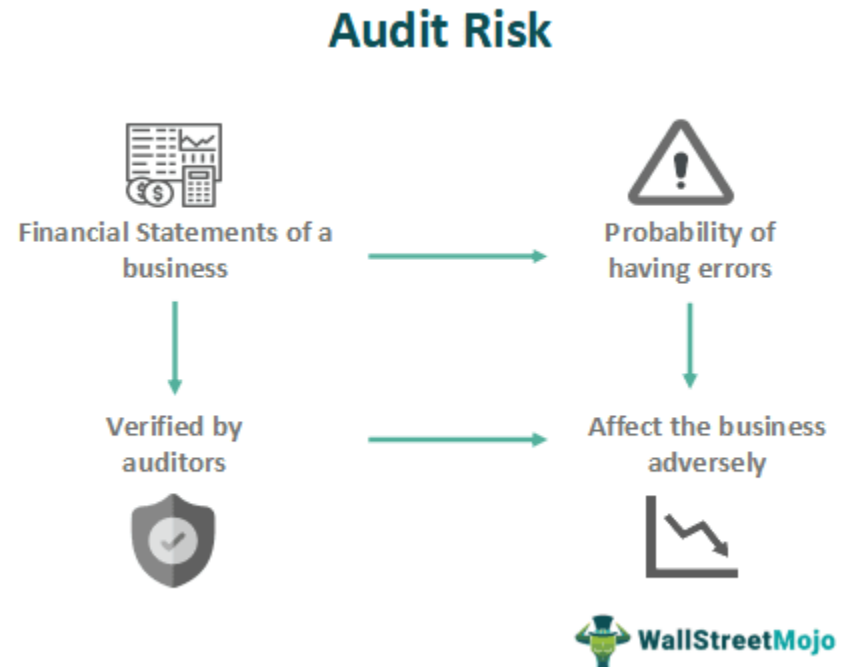
- Reserve analysis compares the amount of the contingency reserves remaining to the amount of risk remaining at any time in the project in order to determine if the remaining reserve is adequate.
- Reserve analysis from starting to end of the project will give indications to health of the project and risks
- Remember Critical Path Method, burndown chart



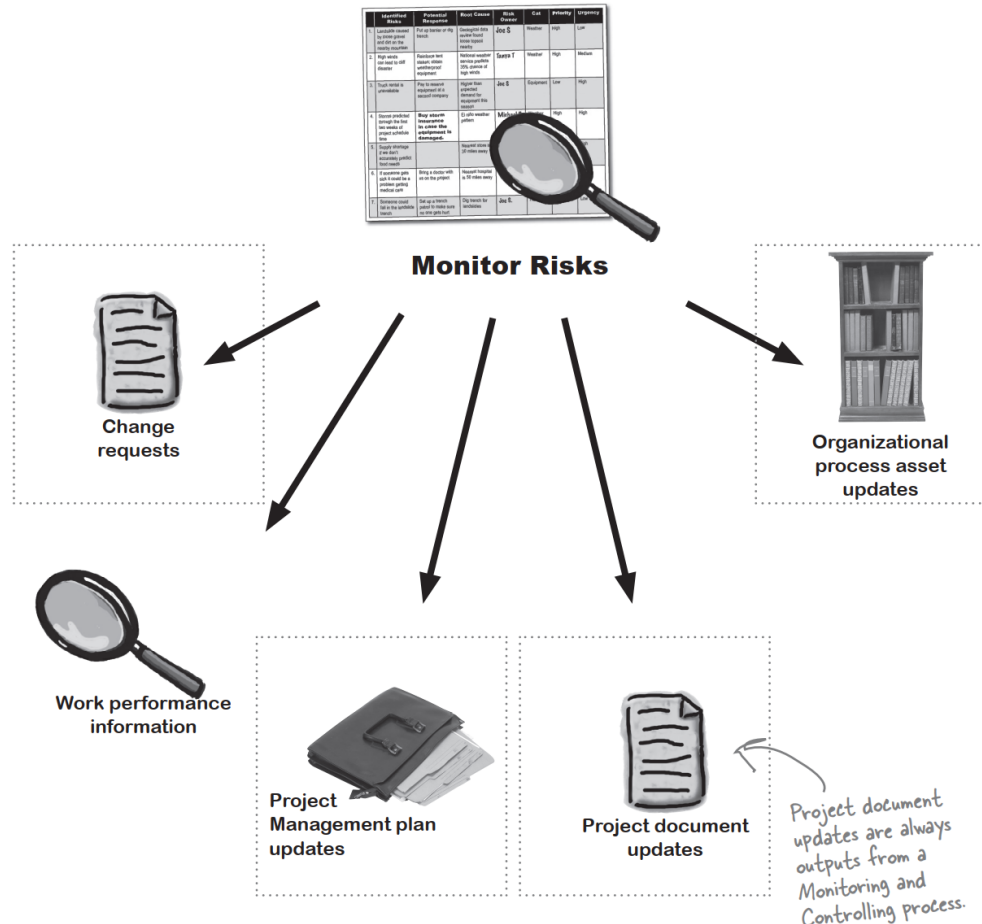
Tools & Techniques

Audits are when you have an outside party come in and take a look at your risk response strategies to judge how effective they are. Sometimes risk audits will point out better ways of handling a specific risk so that you can change your response strategy going forward.

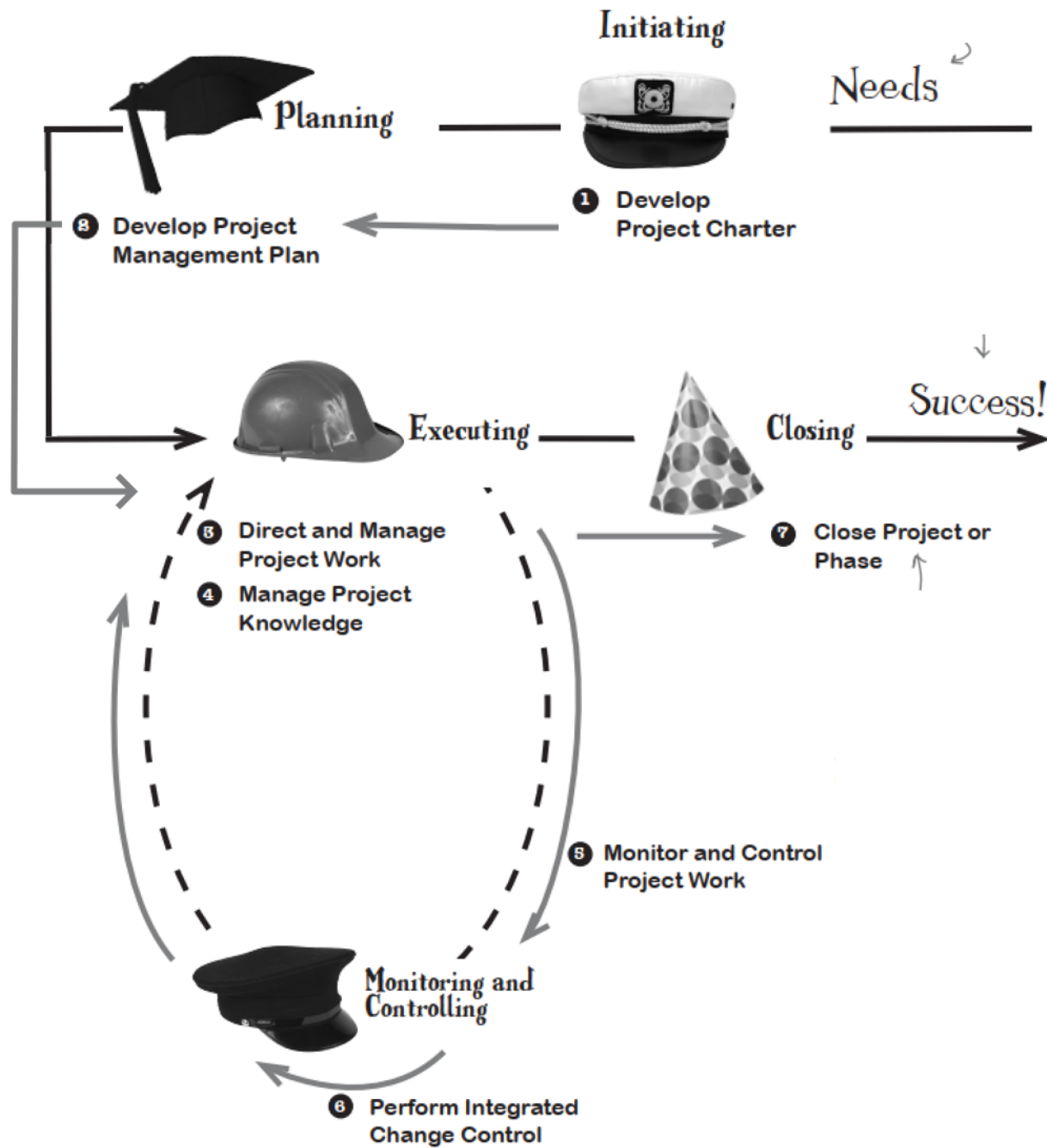
- **Internal:** by the project team, PMO (Project Management Office), or internal audit department.
- **External:** external consultants, auditing firms, or regulatory bodies.



Outputs



Monitor and Control Project
Work process
& Perform Integrated Change
Control process





**Direct and Manage
Project Work**

1

YOU START WITH
INFORMATION ABOUT
HOW THE WORK IS
BEING PERFORMED.

**Monitor
and Control
Project Work**

2

NEXT YOU FIGURE OUT
ANY CHANGES THAT
HAVE TO BE MADE
TO THE PLANS, AND
REPAIRS THAT HAVE
TO BE MADE TO THE
DELIVERABLES.

Here, you let
stakeholders know about
the changes, and make
sure everyone is in the
loop about what you're
doing.

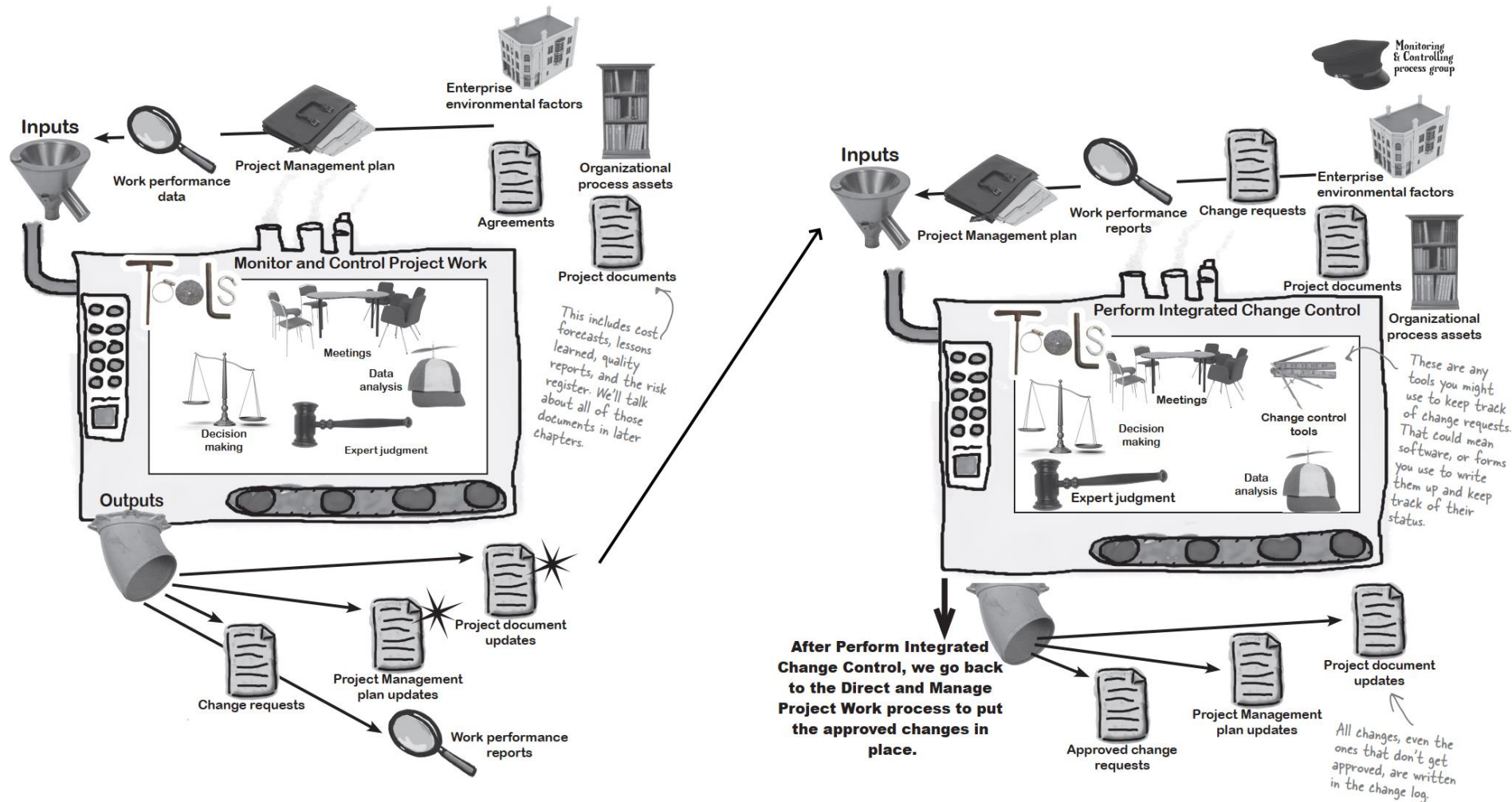
**Perform Integrated
Change Control**

3

ONCE THE
CHANGES AND
REPAIRS ARE
APPROVED, YOU
SEND THEM BACK
TO THE TEAM
TO PUT THEM IN
PLACE.

A change control board (CCB) is a group of people—usually including the sponsor—that approves or rejects changes. Any time a change goes through integrated change control, the CCB decides whether or not it should be made. When they approve the change, you send it on to the team to implement.

Look for changes and deal with them



*Make only the **changes** that are **right** for your project*

Here's where you notice a problem.

**Monitor
and Control
Project Work**

So you write down exactly
what that change should be
and send it to the CCB.



Change request

You use the change control
system to evaluate the
merits of the requested
change and decide to
approve or reject it.



**Perform
Integrated
Change Control**

If the Perform Integrated Change
Control process says it's OK to go
ahead, you make the change. The
important thing to remember is
that the change requests come from
Monitor and Control Project Work,
but are actually made during Direct
and Manage Project Work.



Approval

All of the decisions your CCB
makes about change requests should
be written down in the change log
for the project.



Make the change!

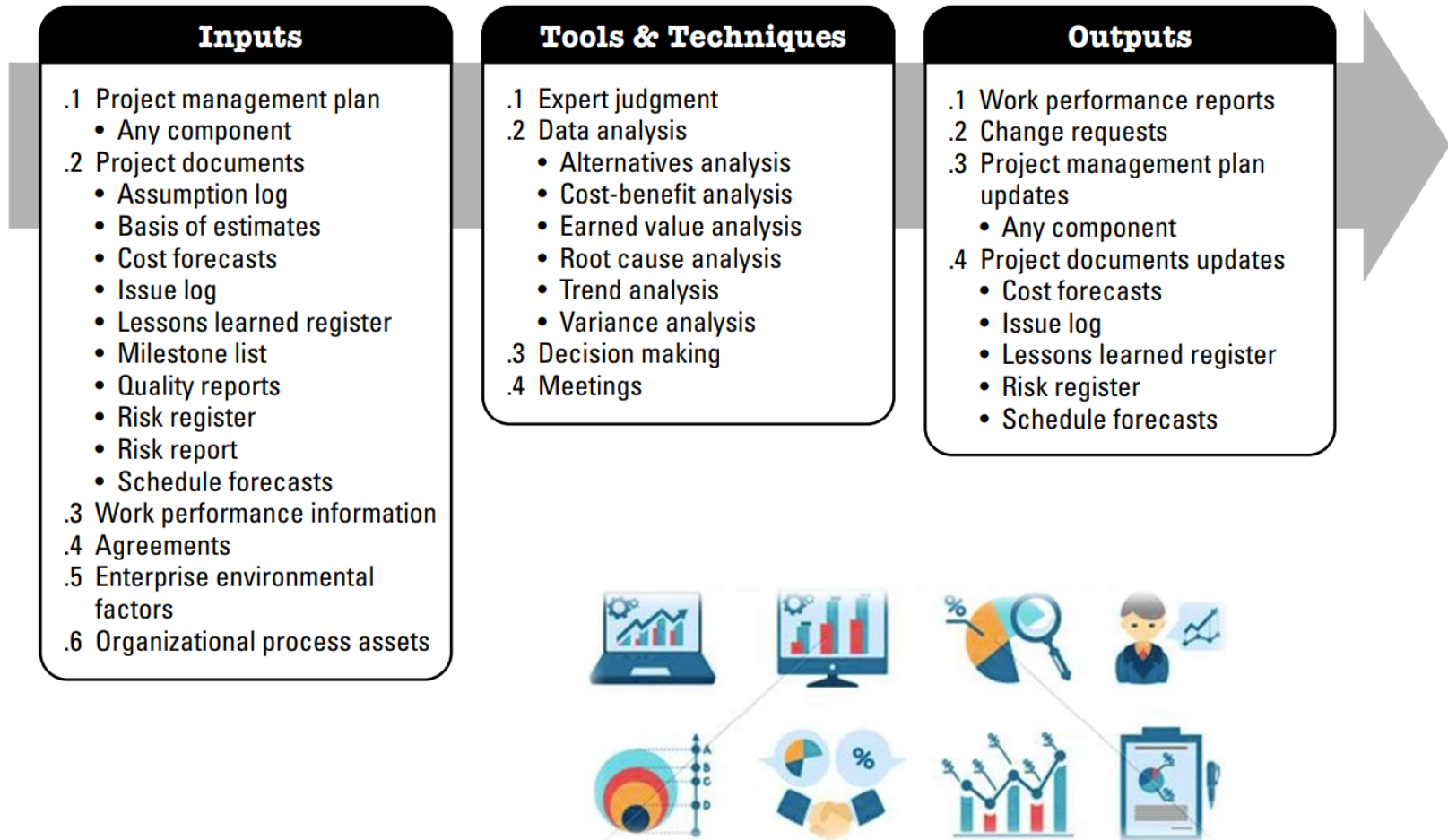
**Direct and Manage
Project Work**



Monitor and Control Project Work

- **What:** process of tracking, reviewing, and reporting the overall progress to meet the performance objectives defined in the project management plan.
- **Why**
 - Allows stakeholders to understand the current state of the project
 - To recognize the actions taken to address any performance issues
 - To have visibility into the future project status with cost and schedule forecasts
- **How:** Monitoring includes collecting, measuring, and assessing **measurements and trends** to effect process improvements.
 - insight into the health of the project
 - identifies any areas that may require special attention

Process



Outputs

Work Performance Reports

- **Work performance information** is combined, recorded, and distributed in a physical or electronic form (**Work performance reports**) in order to create awareness and generate decisions or actions.
- They are circulated to the project stakeholders through the communication processes as defined in the project communications management plan.

Change Requests

- As a result of comparing planned results to actual results

Project Management Plan updates

- Project Management Plan
- Baselines

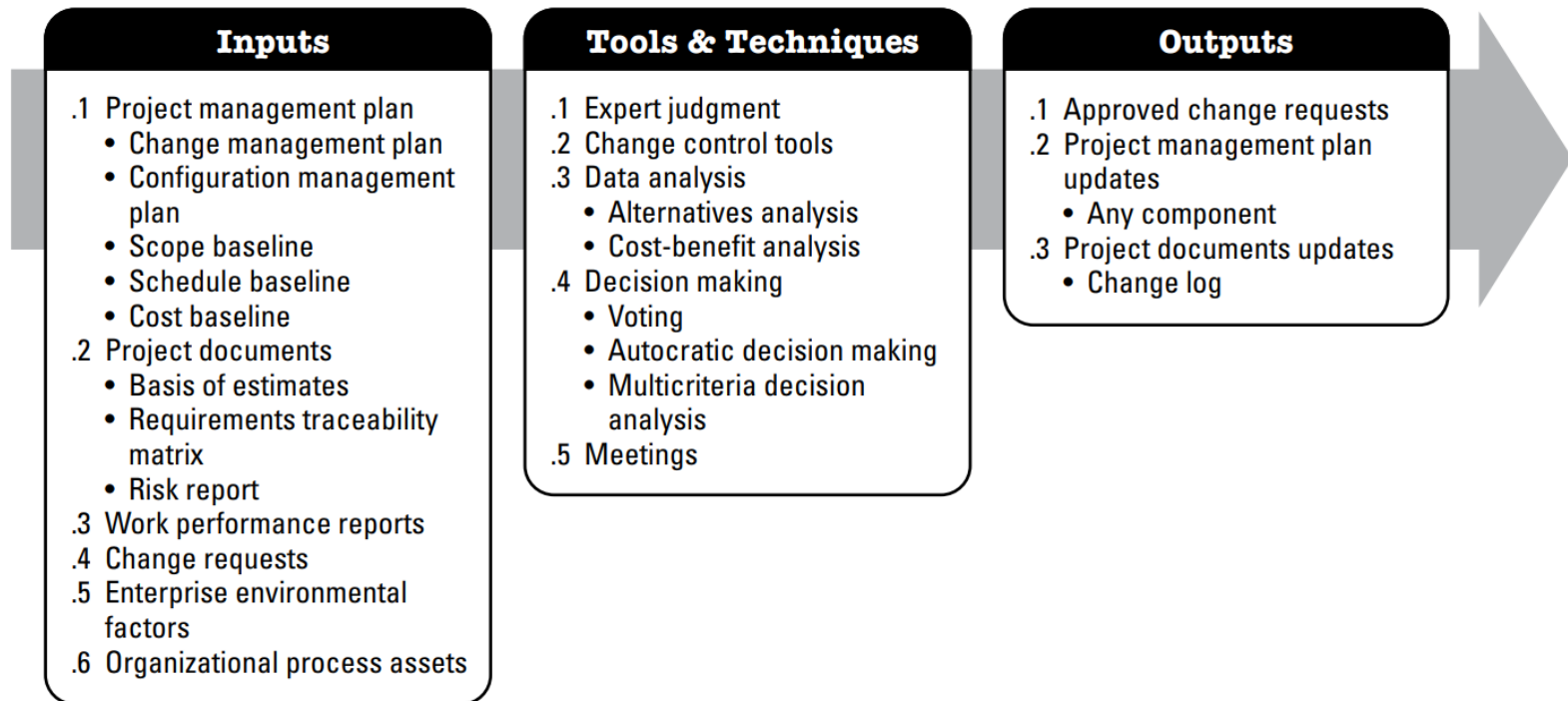
Project documents updates

- **Schedule forecasts.** Changes in schedule forecasts
- **Cost forecasts.** Changes in cost forecasts
- **Issue log.** New issues raised.
- **Lessons learned register.** effective responses for variances and corrective and preventive actions.
- **Risk register.** New risks identified

Perform Integrated Change Control

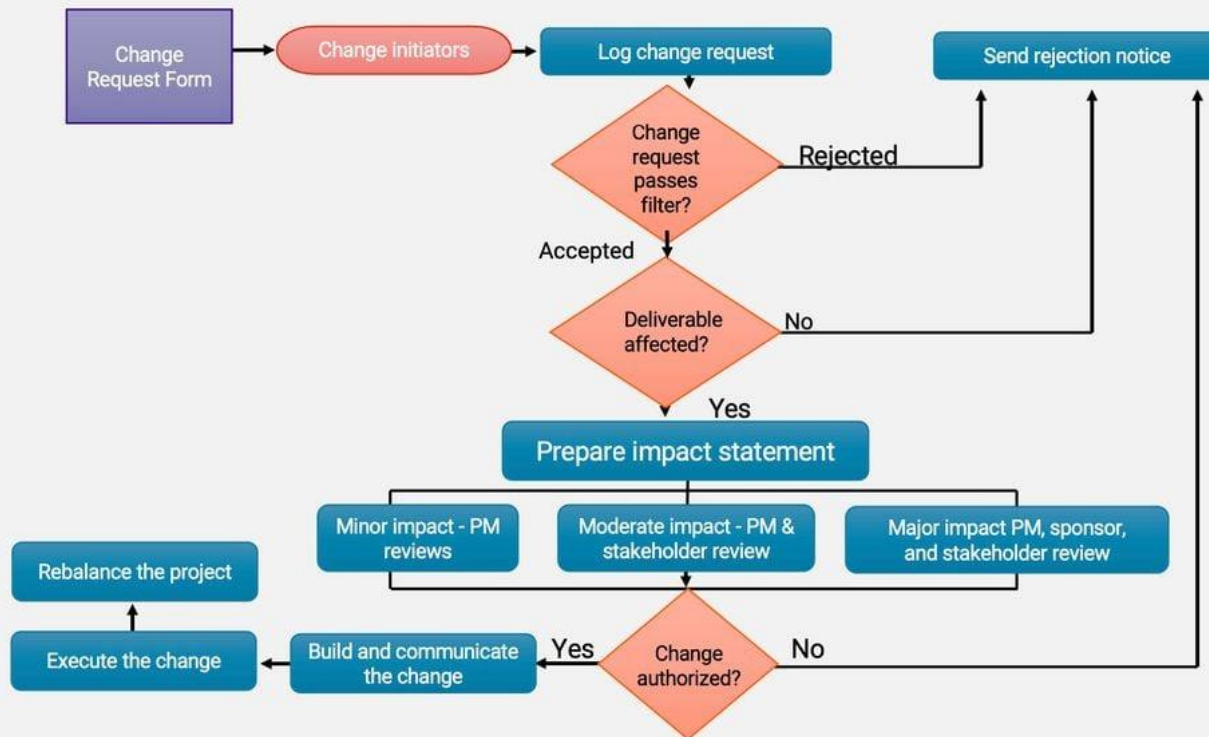
- What
 - Reviewing all change requests
 - Approving changes
 - Managing changes to deliverables, project documents, and the project management plan
 - Communicating the decisions
- Why
 - allows for documented changes within the project to be considered in an **integrated manner** while **addressing overall project risk**

Perform Integrated Change Control



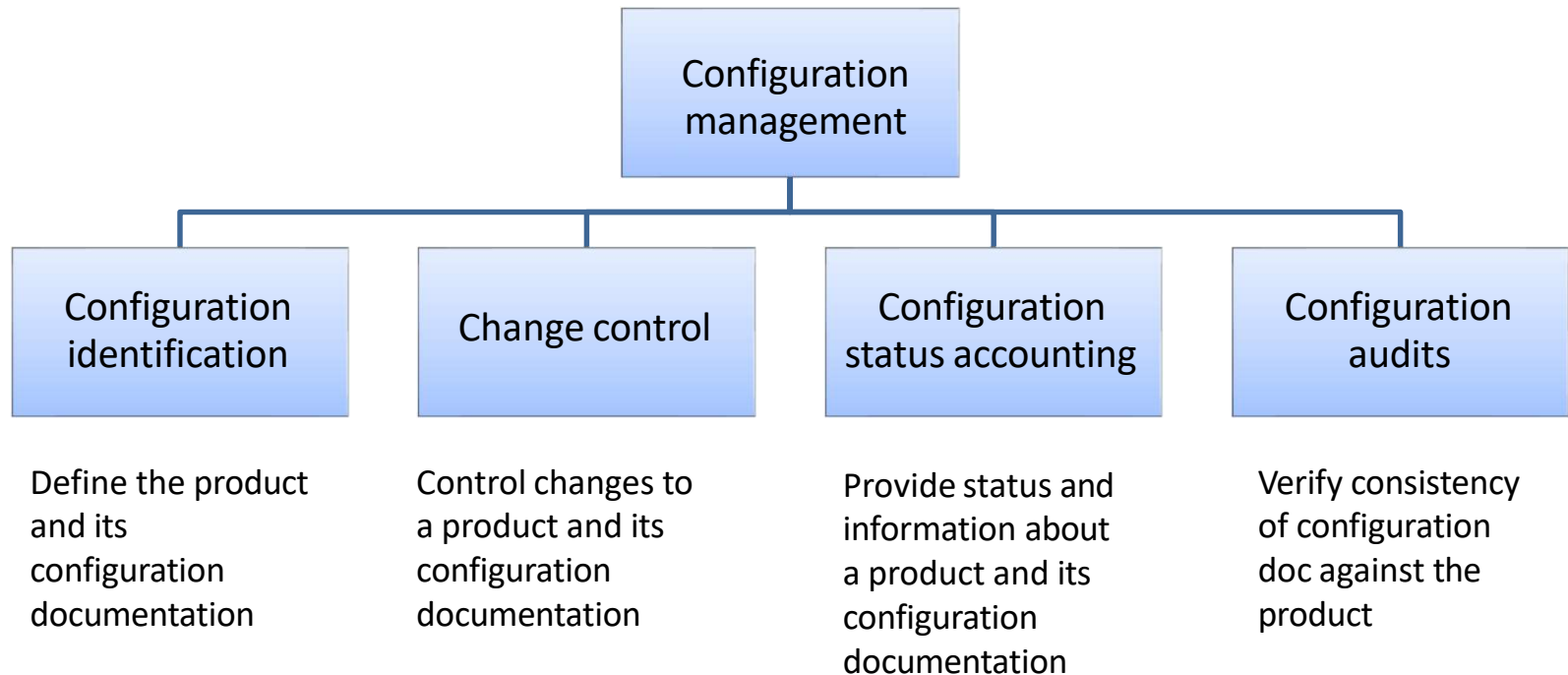
Perform Integrated Change Control

Change Management Process Flowchart



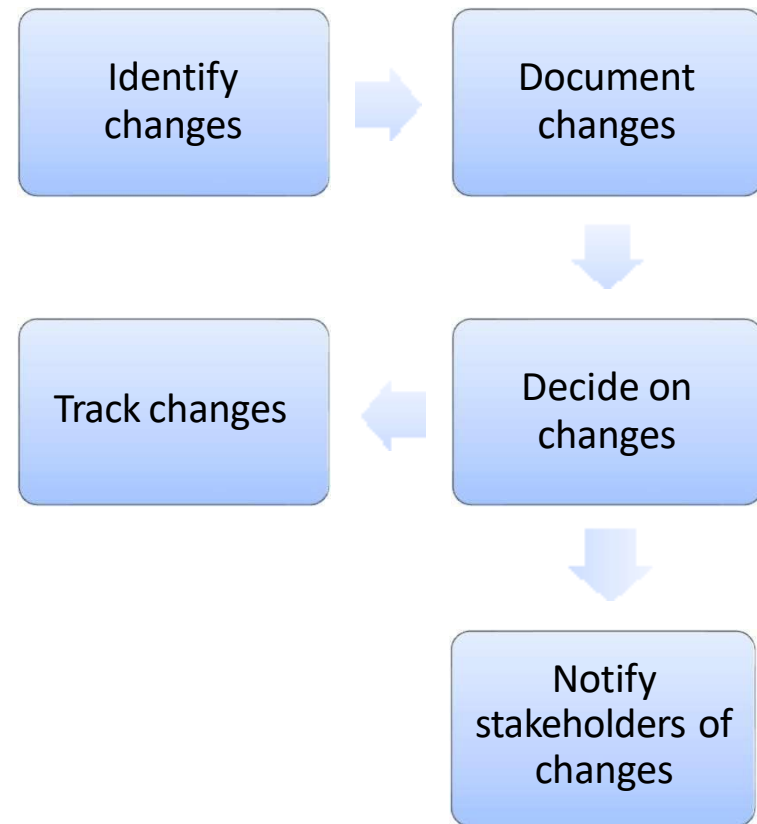
Change control tools

- In order to facilitate configuration and change management, manual or **automated tools** may be used.
- Tools should support the following **configuration management** activities:



Change control tools

- Tools are also used to manage the change requests and the resulting decisions.
- Tools should support the following **change management** activities:
 - Identify changes.
 - Document changes.
 - Decide on changes.
 - Track changes.
 - Notify stakeholders of changes



Tools & Techniques

Alternatives analysis.

- Assess the requested changes and decide which are accepted, rejected, or need to be modified to be finally accepted.

Cost-benefit analysis.

- Determine if the requested change is worth its associated cost.

Decision making

- To decide on whether to accept, defer, or reject change requests.
 - Voting
 - Autocratic decision making
 - Multicriteria decision analysis.

Meetings

- A change control board (CCB) is responsible for meeting and Reviewing the change requests and **approving, deferring or rejecting** those change requests.
- The CCB may meet only once a week, once every other week, or even once a month, depending on the project.
- When emergencies arise, the **pre-established procedures** allow the project manager to implement the change on the spot.
- The CCB may also review configuration management activities.



Outputs

Approved Change Requests

- As a result, changes may be approved, deferred, or rejected.
- **Approved change requests** will be implemented through the Direct and Manage Project Work process.
- **Deferred or rejected change requests** are communicated to the person or group requesting the change.
- The status of all the changes, approved or not, will be updated in the **change log** as part of the project document updates

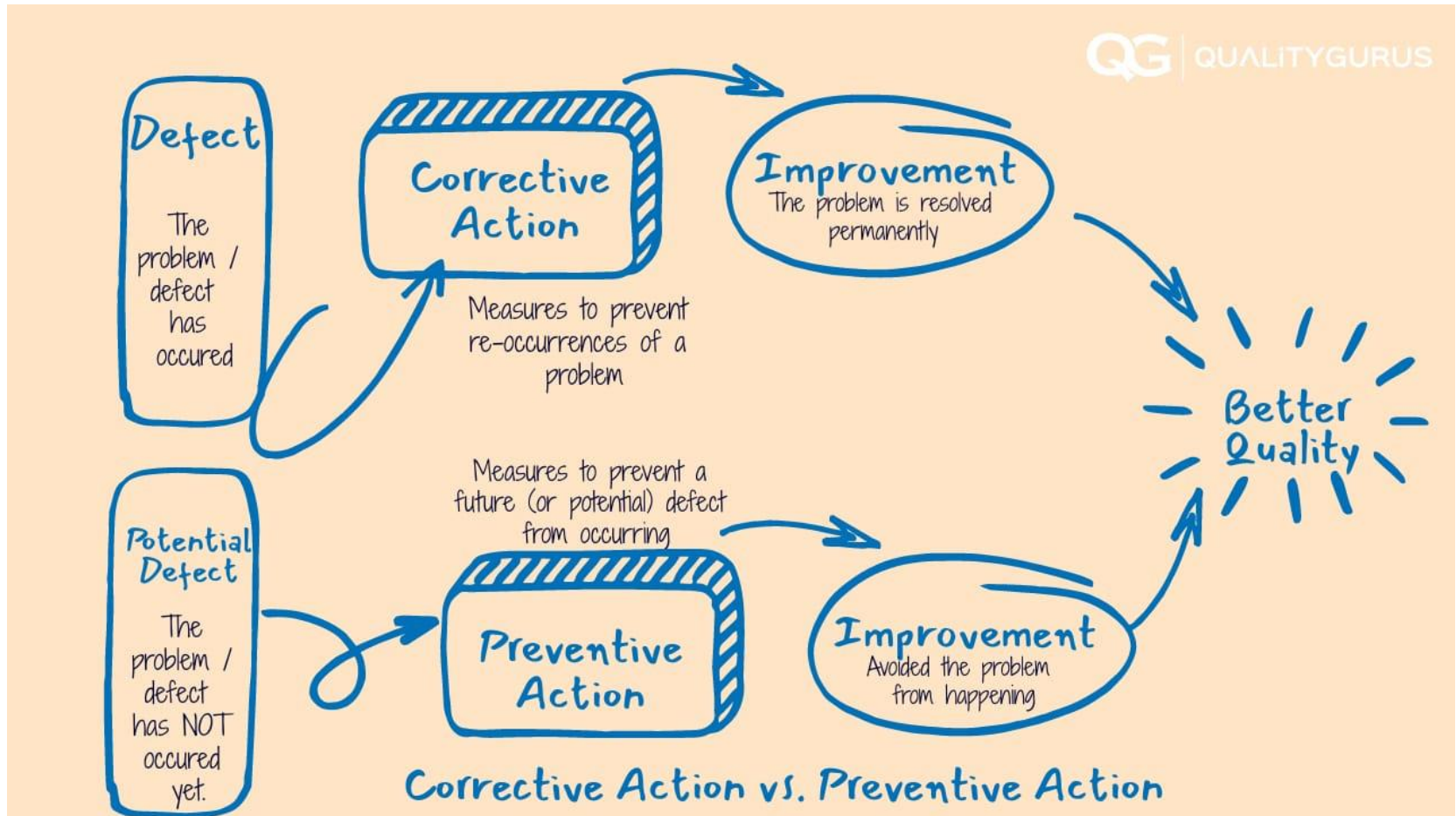
Project Management Plan Updates

- Changes to baselines are only made from the last baseline forward.
- Past performance is not changed.

Project Document Updates

- **Change log:** Log or record of changes made to a project
- Any documents that are subject to the formal change control process

Preventing or correcting problems



*In both corrective and preventive action, you always need to **submit your proposed change** and put it through the Perform Integrated Change Control process—and **only if it is approved** will you implement it.*